

University of Moratuwa

Faculty of Information Technology

CM3321 - Logic Programming and Artificial Cognitive Systems

Gemstone Identification Assistant

A Rule-Based Expert System for Identifying Sri Lankan Gemstones and
Detecting Synthetics and Imitations

Project Report

Name: Sansiluni J.A.S

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Live demo (no installation): <https://gemstone-identification-assistant.onrender.com>

Source code (GitHub): <https://github.com/Suweka/Gemstone-Identification-Assistant>

User manual: Appendix D, page 56

Implemented in SWI-Prolog 10.1.16

Abstract

Gem buyers, students and small traders in Sri Lanka frequently struggle to tell a valuable natural gemstone from a cheaper look-alike, a laboratory-grown synthetic or a glass imitation, while professional laboratory testing costs time and money. This project designs and implements the **Gemstone Identification Assistant**, a rule-based expert system written in SWI-Prolog that gives a quick, explained first opinion. The knowledge base contains 297 facts and 62 production rules, acquired from standard gemmological references and organised in four chained layers: species, variety, origin and treatment, and buying recommendation. The inference engine provides three reasoning modes over the same rule base: forward chaining (data-driven identification), backward chaining (goal-driven verification of a claim, implemented as a meta-interpreter that builds proof trees) and an interactive consultation that asks the user only for the facts it needs. MYCIN-style certainty factors express the confidence of each conclusion and turn incomplete data into a ranked list of candidates. A full explanation facility answers *How*, *Why* and *Why not*, both as a technical trace and in plain language, and animated reasoning diagrams show how each chaining method worked for the user's own input. A web interface, a console interface and a natural-language query box built with a Definite Clause Grammar make the system accessible to non-experts. The web interface checks readings as they are typed, uses pictures for colours, optical effects and inclusions, suggests the most useful next measurement when data is incomplete, and lets users print or save any result. Twenty documented test cases and 47 automated unit tests pass. The report closes with a discussion of how the system models expert cognition, a comparison with machine-learning approaches, and an honest account of its limitations.

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1. Introduction and Problem Definition

1.1 Background

Sri Lanka is one of the world's most important sources of coloured gemstones. Sapphire, ruby, spinel, chrysoberyl (including alexandrite and cat's-eye), garnet, zircon, topaz, tourmaline, quartz and moonstone are mined and traded in large numbers, particularly around Ratnapura. The same trade attracts imitations (coloured glass, cubic zirconia), synthetics (laboratory-grown corundum and spinel) and undisclosed treatments such as heating. Gemmologists separate these materials with a small set of physical and optical tests: refractive index, specific gravity, optic character, optical phenomena and the inclusions seen under magnification [1]-[4]. These tests follow well-documented reasoning, which makes gem identification a good candidate for a rule-based expert system.

1.2 Problem statement

Buyers, students and small traders often cannot tell a valuable natural gem from a cheaper look-alike or a fake, and deception is a known problem in the local trade. Accredited laboratories such as the National Gem and Jewellery Authority (NGJA) of Sri Lanka [7] give reliable answers but cost money and time, so they are not practical for every stone a buyer is offered. There is a need for a tool that takes the observations an informed buyer or student can realistically make and returns a quick, **explained** first opinion: what the stone probably is, whether it is likely natural, synthetic or an imitation, how confident that opinion is, and what the buyer should do next.

1.3 Aim and objectives

The aim is to build an expert system that identifies common Sri Lankan gemstones and detects synthetics and imitations, and that explains its reasoning. The objectives are:

1. O1 - Acquire gemmological knowledge from recognised sources and document how it became facts and rules.
2. O2 - Represent the knowledge as Prolog facts and production rules, with at least 20 of each.
3. O3 - Implement both forward and backward chaining over the same rule base.

4. O4 - Provide an explanation facility that shows user input, facts, rules fired, reasoning and conclusion.
5. O5 - Handle incomplete data and express the confidence of conclusions.
6. O6 - Provide a user-friendly interface for non-experts.
7. O7 - Test the system systematically and document it with a user manual.

1.4 What the system specialises in

The system specialises in ten gem species that are common in Sri Lanka (corundum, spinel, garnet, chrysoberyl, zircon, topaz, tourmaline, quartz, orthoclase feldspar and beryl), eighteen named varieties (for example ruby, blue sapphire, padparadscha, alexandrite, cat's-eye chrysoberyl, moonstone, hessonite and aquamarine) and the two most common imitation materials, glass and cubic zirconia. It reasons from six observable properties: refractive index (RI), specific gravity (SG), optic character, body colour, optical phenomenon and the dominant inclusion type.

1.5 Conclusions the system provides

Table 1. Conclusions produced by the system

Conclusion	Description and example
Species	The mineral species, e.g. <i>corundum</i> , from optic character, RI and SG.
Variety	The trade variety within the species, e.g. corundum + blue gives <i>blue sapphire</i> ; chrysoberyl + colour change gives <i>alexandrite</i> .
Imitation	Glass or cubic zirconia, e.g. an isotropic stone with a glass RI and gas bubbles.
Origin and treatment	Likely natural, synthetic (laboratory-grown), imitation, glass-or-synthetic, laboratory test needed, or not determined; and probable heat treatment.
Confidence	A certainty factor for every derived fact (e.g. <i>blue sapphire</i> 90%), a confidence level (strong match / possible match / no match), and a ranked candidate list when data is missing (e.g. <i>Spinel</i> 52%, <i>Garnet</i> 52%).
Recommendation	Practical advice, e.g. " <i>Features are consistent with a natural stone. Obtain an NGJA certificate before purchase.</i> "
Next measurement	When data is incomplete, which missing measurement would best separate the remaining candidates, e.g. " <i>Measure the refractive index next: Spinel 1.712-1.736, Garnet 1.730-1.890.</i> "
Explanation	How the conclusion was reached, why a question is being asked, and why an alternative was rejected.

1.6 Scope and limitations

The system is a decision-support tool for a first opinion; it is **not** a replacement for laboratory certification. The knowledge base covers ten species and two imitations. Heat treatment can only be suggested from a visible indicator (discoid fractures), not proven. The system assumes the user can take basic readings (a refractometer, a hydrostatic balance and a polariscope) but works with partial data. Section 7.3 discusses the limitations in detail.

2. Knowledge Engineering

Knowledge engineering was carried out as an iterative process (Figure 1). The domain was studied from recognised sources, individual knowledge statements were extracted into a knowledge acquisition table, encoded as Prolog facts and rules, tested, and refined when tests revealed gaps.

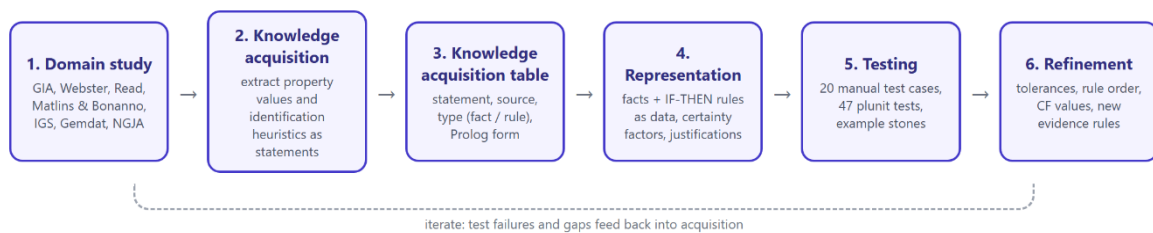


Figure 1. Knowledge engineering process used in the project

2.1 Domain study: gemmology concepts

Refractive index (RI) measures how strongly a gem slows and bends light. It is read on a gem refractometer and is the single most diagnostic test for most transparent gems [3]. Corundum reads about 1.762-1.770, quartz about 1.544-1.553 [1], [3]. Standard refractometers cannot read above about 1.81, so very high-RI materials such as high zircon and cubic zirconia show as "over the limit" (no shadow edge on the scale). This observation is itself useful evidence, and the system accepts it as an input.

Specific gravity (SG) is the density of the stone relative to water, measured by hydrostatic weighing. It separates gems with similar RI; corundum (about 4.00) is much heavier than spinel (about 3.60) [2].

Optic character describes how light passes through the crystal. Isotropic (singly refractive) materials such as spinel, garnet, glass and cubic zirconia transmit light as a single ray. Anisotropic (doubly refractive) materials split it in two; they are *uniaxial* (one optic axis:

corundum, quartz, beryl, tourmaline, zircon) or *biaxial* (two axes: chrysoberyl, topaz, feldspar) [2], [3]. It is determined with a polariscope and conoscope. **Birefringence** is the difference between the two RI readings of a doubly refractive gem.

Optical phenomena are special light effects: asterism (a star) and chatoyancy (a cat's-eye) caused by aligned needle-like inclusions, colour change (typical of alexandrite) and adularescence (the floating sheen of moonstone) [1].

Inclusions are the gemmologist's main field evidence for origin. Rutile "silk" and mineral crystals indicate natural growth; curved growth lines (striae) and round gas bubbles indicate flame-fusion (Verneuil) synthetics; swirl marks and bubbles indicate glass; discoid fractures around crystals indicate heat treatment [1], [4], [5].

Synthetic versus imitation. A synthetic has the same chemistry and properties as the natural gem but was grown in a laboratory; an imitation merely looks similar and has different properties [4]. The distinction matters because a synthetic ruby is still corundum and passes the species tests, so only inclusion evidence can reveal it.

Table 2 summarises the property facts acquired for each material. These are refractometer and hydrostatic *reading ranges*, which are slightly wider than textbook mean values so that normal variation within a species is covered.

Table 2. Property facts held in the knowledge base (generated from knowledge_base.pl)

Material	RI range	SG range	Optic character	Birefringence	Hardness
Corundum	1.757 - 1.780	3.95 - 4.10	uniaxial	0.008	9
Spinel	1.712 - 1.736	3.55 - 3.70	isotropic	-	8
Garnet	1.730 - 1.890	3.57 - 4.30	isotropic	-	7
Chrysoberyl	1.740 - 1.760	3.70 - 3.78	biaxial	0.009	8.5
Zircon	1.810 - 1.990	4.60 - 4.80	uniaxial	0.059	7.5
Topaz	1.606 - 1.644	3.49 - 3.57	biaxial	0.009	8
Tourmaline	1.614 - 1.666	3.00 - 3.26	uniaxial	0.018	7.5
Quartz	1.540 - 1.558	2.60 - 2.70	uniaxial	0.009	7
Feldspar	1.515 - 1.530	2.55 - 2.63	biaxial	0.006	6
Beryl	1.564 - 1.602	2.63 - 2.92	uniaxial	0.006	7.5

Material	RI range	SG range	Optic character	Birefringence	Hardness
Glass	1.440 - 1.770	2.20 - 4.50	isotropic	-	5.5
Cubic zirconia	2.150 - 2.180	5.60 - 6.00	isotropic	-	8.5

2.2 Domain study: expert system concepts and related work

An expert system captures the specialist knowledge of human experts so that non-experts can apply it [10], [11]. The classic architecture separates a **knowledge base** (domain facts and heuristics), an **inference engine** (a domain-independent reasoning procedure), **working memory** (the facts of the current case) and a **user interface** with an **explanation facility** [11]. This separation means the knowledge can be extended without changing the reasoning code.

Production rules (IF conditions THEN conclusion) are the most widely used knowledge representation in expert systems because they match the way experts state heuristics and because each rule can be explained individually [10]. Two inference strategies operate on production rules [11], [17]:

- **Forward chaining** is data-driven. Starting from the known facts, the engine repeatedly finds a rule whose conditions are all satisfied, adds its conclusion to working memory, and stops when no rule can add anything new. When several rules are applicable, a **conflict resolution** strategy (for example rule order or specificity) chooses which fires, and **refractoriness** prevents the same rule instance firing twice.
- **Backward chaining** is goal-driven. Starting from a hypothesis, the engine looks for rules that conclude it and tries to prove their conditions as sub-goals, recursively, until it reaches known facts. Prolog's own execution model is backward chaining with depth-first search and backtracking [12].

The best-known rule-based expert system, **MYCIN**, diagnosed bacterial infections using about 450 backward-chained rules [8]. It introduced two ideas this project adopts. The first is **certainty factors**: each rule carries a measure of belief, and evidence from several rules is combined incrementally [9]. The second is a consultation dialogue that asks the user only for the information a rule needs and can answer "why?" by showing the rule being tried [8], [9]. Later work such as XPLAIN argued that good explanations must justify the reasoning, not merely replay it [20]. That motivated the plain-language explanations and the source-linked rule justifications in this system.

Bratko [12] and Merritt [13] show how to build expert system shells in Prolog: rules are stored as data, a meta-interpreter proves goals and records a proof tree for "how" explanations, and questions are asked lazily during proof. Sterling and Shapiro [14] describe the meta-interpreter technique in depth, and Pereira and Warren [15] introduced Definite Clause Grammars (DCGs), which this project uses for its natural-language interface. SWI-Prolog [16] provides the HTTP and HTML libraries used for the web interface.

Related work on gemstone identification. Most published work on gem identification concentrates on laboratory instruments. Fritsch and Stockton [21] showed how infrared spectroscopy can simplify identification by comparing a stone's spectrum with a library of standard reference spectra. Jenkins and Larsen [23] described identification with Raman spectroscopy, noting that gems are usually examined by trained personnel using a variety of methods. Karampelas et al. [22] review the instruments that help to meet the challenges of the modern gem market, where treatments are widespread and the origin of a gem has become more important than in the past. These methods are accurate, but they need laboratory equipment and specialist training that a buyer in a local market rarely has. More recently, Raheem et al. [24] used convolutional neural networks to improve the accuracy of gemstone identification from images. This is the machine-learning direction discussed in Section 7.2: such models can be accurate on photographs, but they do not explain their decisions and cannot use RI, SG or optic character.

Related work on rule-based expert systems. Buchanan and Duda [25] surveyed the principles of rule-based expert systems and the production-system model on which this project is built. Grosan and Abraham [26] and Abraham [27] describe the architecture of such systems, including the explanation facility that allows a user to understand how the system arrived at a conclusion. Clancey [28], working from MYCIN, asked how domain-independent strategic knowledge can be represented in a rule-based system so that its reasoning can be explained and taught, which supports this project's emphasis on justifying every conclusion.

How this system differs. Compared with the instrumental studies [21]-[23], the Gemstone Identification Assistant works from the basic readings and observations available to a buyer or student (a refractometer, a hydrostatic balance, a polariscope and a loupe) and gives a first opinion rather than a laboratory result. Compared with image-based classification [24], every conclusion is explained step by step and linked to a published source. Building on the rule-based principles in [25]-[28], it adds a focus on Sri Lankan gems and the local trade, three reasoning modes over one rule base (forward chaining, backward chaining and an interactive

consultation), certainty factors that turn incomplete data into ranked candidates with a suggested next measurement, and plain-language How, Why and Why-not explanations with reasoning diagrams for non-experts.

2.3 Knowledge acquisition

Knowledge was acquired from documentary sources: the GIA Gem Encyclopedia [1], Webster's *Gems* [2], Read's *Gemmology* [3], Matlins and Bonanno's *Gem Identification Made Easy* [4], the International Gem Society [5], the Gemdat database [6] and the NGJA [7]. Each usable piece of knowledge was written as a short statement, classified as a fact or a rule, and encoded in Prolog. Every rule in the knowledge base stores its justification and source in `rule_info(Id, Justification, Source)`, so the acquisition trail is kept inside the system itself and appears in every explanation. Table 3 shows a representative selection of the 41 main knowledge statements; the complete rule base with justifications is in Appendix A.

Four knowledge engineering decisions were made during acquisition:

1. **Reading ranges plus tolerance.** Published ranges were stored as reading ranges, and the engine adds a measurement tolerance of ± 0.005 for RI and ± 0.03 for SG to allow for instrument and operator error. A reading that fits only because of the tolerance is accepted with reduced certainty (Section 2.5).
2. **Layered rules.** Knowledge was organised by the question it answers (species, variety, origin, advice), so that conclusions of one layer become conditions of the next.
3. **Explicit uncertainty.** Where sources describe evidence as typical rather than conclusive (for example, fingerprints usually indicate a natural stone, but flux-grown synthetics show similar features), the rule was given a lower certainty factor rather than being omitted.
4. **"Over the limit" as a reading.** When a refractometer shows no shadow edge, the only thing known is that the RI is above the instrument's limit (about 1.81). This is stored as the value `ri(over_limit)`, and it matches any material whose RI range extends above the limit (zircon, cubic zirconia, some garnets) with a reduced certainty of 0.85, because the exact value is unknown.

Table 3. Knowledge acquisition table (representative entries; KE = knowledge engineer decision)

#	Knowledge statement (from source)	Source	Type	Prolog form
K1	Corundum reads about 1.762-1.770 on the refractometer.	[1]	Fact	ri_range(corundum, 1.757, 1.780).
K2	Corundum has SG about 4.00.	[1]	Fact	sg_range(corundum, 3.95, 4.10).
K3	Corundum is doubly refractive and uniaxial.	[1]	Fact	optic(corundum, uniaxial).
K4	Spinel is singly refractive, RI about 1.718, SG about 3.60.	[2]	Fact	optic(spinel, isotropic). ri_range(spinel, 1.712, 1.736).
K5	Garnets are singly refractive, RI 1.73-1.89, SG 3.6-4.3 depending on type.	[2]	Fact	ri_range(garnet, 1.730, 1.890).
K6	Chrysoberyl is biaxial, RI about 1.746-1.755, SG about 3.73.	[1]	Fact	optic(chrysoberyl, biaxial).
K7	High zircon is uniaxial with very high RI and SG about 4.7.	[6]	Fact	sg_range(zircon, 4.60, 4.80).
K8	Topaz is biaxial, RI about 1.61-1.64, SG about 3.53.	[1]	Fact	ri_range(topaz, 1.606, 1.644).
K9	Quartz is uniaxial, RI 1.544-1.553, SG 2.65.	[3]	Fact	sg_range(quartz, 2.60, 2.70).
K10	Moonstone (orthoclase) is biaxial, RI about 1.52, SG about 2.57.	[3]	Fact	ri_range(feldspar, 1.515, 1.530).
K11	Beryl is uniaxial, RI about 1.57-1.60, SG about 2.7.	[6]	Fact	optic(beryl, uniaxial).
K12	Glass is singly refractive with a wide range of RI.	[5]	Fact	ri_range(glass, 1.440, 1.770).
K13	Cubic zirconia is isotropic with SG 5.6-6.0.	[1]	Fact	sg_range(cubic_zirconia, 5.60, 6.00).
K14	Corundum occurs in red, pink, orange, yellow, green, blue, purple and colourless.	[1]	Fact	colour(corundum, blue).
K15	Chrysoberyl can show chatoyancy and colour change.	[1]	Fact	phenomenon(chrysoberyl, colour_change).
K16	Readings vary slightly with instrument and operator.	KE	Fact	tolerance(ri, 0.005). tolerance(sg, 0.03).

#	Knowledge statement (from source)	Source	Type	Prolog form
K17	A uniaxial stone with corundum RI and SG is corundum.	[1]	Rule	rule(r1, [optic(uniaxial), ri_in(corundum), sg_in(corundum)], species(corundum)).
K18	A singly refractive stone with RI about 1.72 and SG about 3.60 is spinel.	[2]	Rule	rule(r2, [optic(isotropic), ri_in(spinel), sg_in(spinel)], species(spinel)).
K19	An isotropic stone with a glass RI and gas bubbles that matches no species is glass.	[5]	Rule	rule(r11, [optic(isotropic), ri_in(glass), inclusion(gas_bubbles), not(species(_))], imitation(glass)).
K20	Swirl marks are typical of moulded glass.	[4]	Rule	rule(r12, [inclusion(swirl_marks), not(species(_))], imitation(glass)).
K21	An isotropic stone with SG 5.6-6.0 is cubic zirconia.	[1]	Rule	rule(r13, [optic(isotropic), sg_in(cubic_zirconia)], imitation(cubic_zirconia)).
K22	When data is missing, every gem consistent with all supplied properties remains possible.	KE	Rule	rule(r14, [partial_data, ..., optic_ok(G), ri_ok(G), sg_ok(G), colour_ok(G), phenomenon_ok(G)], candidate(G)).
K23	A matching RI is the strongest single test.	[3]	Rule	rule(r16, [candidate(G), not(missing(ri)), ri_ok(G)], candidate(G)).
K24	Red corundum is ruby.	[1]	Rule	rule(r21, [species(corundum), observed_colour(red), not(phenomenon(star))], variety(ruby)).
K25	Blue corundum is blue sapphire.	[1]	Rule	rule(r24, [species(corundum), observed_colour(blue), not(phenomenon(star))], variety(blue_sapphire)).
K26	Non-red corundum with asterism is star sapphire.	[1]	Rule	rule(r23, [species(corundum), phenomenon(star), not(observed_colour(red))], variety(star_sapphire)).
K27	Pinkish-orange Sri Lankan corundum is traded as padparadscha.	[1]	Rule	rule(r26, [species(corundum), observed_colour(orange), ...], variety(padparadscha)).

#	Knowledge statement (from source)	Source	Type	Prolog form
K28	Chrysoberyl with a daylight/incandescent colour change is alexandrite.	[1]	Rule	rule(r29, [species(chrysoberyl), phenomenon(colour_change)], variety(alexandrite)).
K29	Orthoclase feldspar showing adularescence is moonstone.	[1]	Rule	rule(r31, [species(feldspar), phenomenon(adularescence)], variety(moonstone)).
K30	Orange grossular garnet is hessonite.	[1]	Rule	rule(r34, [species(garnet), observed_colour(orange)], variety(hessonite)).
K31	Curved striae in corundum prove flame-fusion synthesis; natural corundum grows in straight lines.	[1]	Rule	rule(r41, [species(corundum), inclusion(curved_lines)], origin(synthetic)).
K32	Round gas bubbles in corundum indicate a flame-fusion synthetic.	[1]	Rule	rule(r42, [species(corundum), inclusion(gas_bubbles)], origin(synthetic)).
K33	Gas bubbles in an unidentified stone indicate glass or a synthetic.	[5]	Rule	rule(r44, [inclusion(gas_bubbles), not(species(_)), not(imitation(_))], origin(glass_or_synthetic)).
K34	Rutile silk is a natural inclusion in corundum.	[1]	Rule	rule(r46, [species(corundum), inclusion(silk)], origin(likely_natural)).
K35	Included mineral crystals indicate natural growth.	[4]	Rule	rule(r47, [species(_), inclusion(crystals)], origin(likely_natural)).
K36	Fingerprints are usually natural, but flux synthetics show similar features.	[4]	Rule (CF 0.6)	rule(r48, [species(_), inclusion(fingerprints)], origin(likely_natural)).
K37	Discoïd fractures form when corundum is heated.	[1]	Rule	rule(r49, [species(corundum), inclusion(discoïd_fractures)], treatment(likely_heated)).
K38	Natural corundum is rarely clean; a clean stone needs a laboratory test.	[5]	Rule	rule(r51, [species(corundum), inclusion(none)], origin(needs_lab_test)).
K39	Natural stones should be certified before purchase (NGJA).	[7]	Rule	rule(r61, [origin(likely_natural)], advice(get_certificate)).
K40	Treatments must be disclosed to the buyer.	[7]	Rule	rule(r67, [treatment(likely_heated)], advice(disclose_heat)).

#	Knowledge statement (from source)	Source	Type	Prolog form
K41	Alexandrite must be confirmed under daylight and incandescent light.	[1]	Rule	rule(r68, [variety(alexandrite)], advice(check_colour_change)).

2.4 Knowledge representation

The knowledge is represented as **Prolog facts plus production rules**. Prolog suits this because it is declarative (facts and rules are written as logical statements), has pattern matching through unification, and has built-in backtracking, which is already a form of backward chaining [12].

The key design decision is that every rule is stored as a **data term**, `rule(Id, Conditions, Conclusion)`, rather than as a Prolog clause. This has three benefits. The same rule base can be interpreted by both a forward chainer and a backward chainer. Every rule has an identifier that can be shown in explanations. Rules can be inspected, listed on the knowledge base page and exported for documentation. Conditions are either facts in working memory (for example `observed_colour(red)`), conclusions of other rules (for example `species(corundum)`), computed tests (for example `ri_in(corundum)`, which compares the RI reading with the stored range), or negations `not(C)` evaluated by negation as failure.

Numeric ranges are kept in facts rather than inside rules: rule `r1` states `ri_in(corundum)` and the engine looks up `ri_range(corundum, Min, Max)`. This keeps the property data in one place and lets the explanation say "RI 1.765 lies within the corundum range 1.757-1.780". Table 4 documents the predicates.

Table 4. Main predicates of the knowledge base and working memory

Predicate	Meaning
<code>gem(Name)</code>	A gem species known to the system (10).
<code>imitation_material(Name)</code>	An imitation material (glass, cubic zirconia).
<code>ri_range(Gem, Min, Max)</code>	Refractive index reading range.
<code>sg_range(Gem, Min, Max)</code>	Specific gravity range.
<code>optic(Gem, Type)</code>	Optic character: isotropic, uniaxial or biaxial.
<code>hardness(Gem, H), birefringence(Gem, B)</code>	Mohs hardness and birefringence (information only).
<code>colour(Gem, C), typical_colour(Gem, C)</code>	Possible colours; colours most common in the trade.

Predicate	Meaning
phenomenon(Gem, P)	Optical phenomena the species can show.
inclusion_type(I, Description)	Inclusion types the user can report.
rule(Id, Conditions, Conclusion)	Production rule (62).
rule_cf(Id, CF)	Certainty factor of a rule.
rule_info(Id, Text, Source), source(Id, Citation)	Justification and literature source of each rule.
advice_text(Key, Text)	Wording of each recommendation.
fact(F)	Working memory: observed and derived facts of the current case (dynamic, thread-local).
fired(Id, Conds, Concl)	Log of fired rules, used for the How explanation.
cf(F, CF), fired_cf/6	Certainty of each derived fact and of each rule firing.

The rules are grouped into four layers (Figure 2). Layer 1 identifies the species from the three core measurements, detects imitations, and, when measurements are missing, produces ranked candidates. Layer 2 names the variety. Layer 3 judges origin and treatment from inclusions. Layer 4 turns conclusions into advice. Because each layer uses the conclusions of the one above as conditions, a single identification typically chains three or four rules.

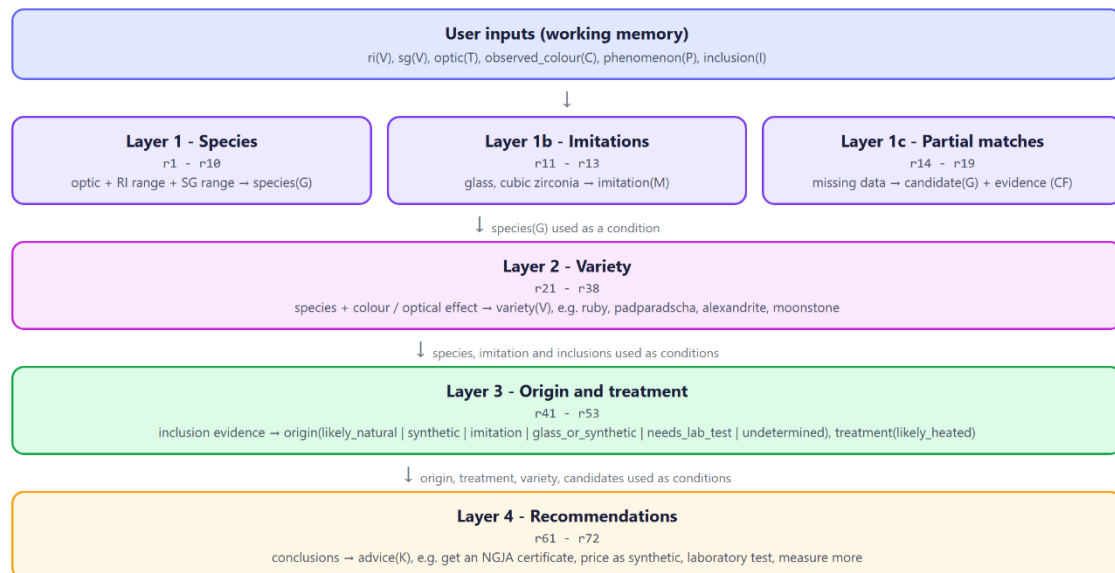


Figure 2. Rule layers and how conclusions chain between them

How to read a rule. Every rule has the form `rule(Id, [Condition1, Condition2, ...], Conclusion)`. The list in square brackets holds the IF part, and all its conditions must be true (they are joined by AND). The last argument is the THEN part: the new fact the rule adds. `not(C)` means "C has not been established" (negation as failure). Conditions such as `ri_in(corundum)` are tests that compare the user's reading with the stored range, so rule r1 below reads: *IF the stone is uniaxial AND its RI is in the corundum range AND its SG is in the corundum range, THEN it is corundum*. Table 5 shows one rule from each layer with its meaning; all 62 rules are listed in Appendix A, with a key to the condition terms.

Table 5. Representative rules from each layer, with their meaning

Layer	Rule in the knowledge base	Meaning in plain English
1. Species	<code>rule(r1, [optic(uniaxial), ri_in(corundum), sg_in(corundum)], species(corundum))</code> .	IF the stone is uniaxial AND its RI is in the corundum range AND its SG is in the corundum range, THEN the species is corundum.
2. Variety	<code>rule(r24, [species(corundum), observed_colour(blue), not(phenomenon(star))], variety(blue_sapphire))</code> .	IF the stone is corundum AND it is blue AND it does not show a star, THEN it is a blue sapphire.
3. Origin	<code>rule(r46, [species(corundum), inclusion(silk)], origin(likely_natural))</code> .	IF the stone is corundum AND it contains rutile silk, THEN it is likely natural.
4. Advice	<code>rule(r61, [origin(likely_natural)], advice(get_certificate))</code> .	IF the stone is likely natural, THEN advise the buyer to obtain a laboratory certificate.

Alternative representations were considered. A **decision tree** would force a fixed order of questions and could not use whatever subset of data the user has. **Frames** would describe each gem well but do not naturally express the heuristic IF-THEN knowledge about inclusions and origin. Production rules express both kinds of knowledge, support both chaining directions and give natural explanations, so they were chosen.

2.5 Representing uncertainty: certainty factors

Gem identification is rarely certain: ranges of different species overlap slightly, colour names are judged by eye, and some inclusions occur in both natural and synthetic stones. Following MYCIN [9], each rule has a certainty factor between 0 (no belief) and 1 (certain), stored as `rule_cf(Id, CF)`. The certainty of a conclusion is the rule's certainty multiplied by that of its weakest condition:

$$CF(\text{conclusion}) = CF(\text{rule}) \times \min(CF(\text{condition}_1), \dots, CF(\text{condition}_n))$$

User inputs have CF 1.0. A reading inside a gem's published range has CF 1.0, while a reading that fits only because of the measurement tolerance has CF 0.7. When a second rule supports a fact that is already believed, the two certainties are combined:

$$CF_{\text{combined}} = CF_1 + CF_2 \times (1 - CF_1)$$

Table 6 summarises the values chosen for each rule group; individual values are listed in Appendix A.

Table 6. Certainty factor ranges by rule group

Rule group	Rules	CF range
Species (layer 1)	r1 - r10	0.90 - 0.95
Imitations (layer 1b)	r11 - r13	0.80 - 0.95
Partial-match base and evidence (layer 1c)	r14 - r19	0.15 - 0.50
Varieties (layer 2)	r21 - r38	0.70 - 0.95
Origin and treatment (layer 3)	r41 - r53	0.60 - 1.00
Recommendations (layer 4)	r61 - r72	1.00 - 1.00

The values encode the knowledge engineer's judgement of how conclusive each piece of evidence is. Species rules are high (0.90-0.95) because three independent measurements agree, but not 1.0 because ranges can overlap. Colour-based varieties are lower where the colour boundary is subjective (padparadscha 0.70). Origin rules range from 0.95 for curved striae, which the literature treats as diagnostic of flame-fusion synthesis [1], down to 0.60 for fingerprints, which can also occur in flux-grown synthetics [4]. The partial-match evidence rules are deliberately weak (0.15-0.50) so that incomplete data never produces a falsely confident answer.

3. System Design and Implementation

3.1 Architecture

The system follows the classic expert system architecture (Figure 3). The knowledge base (knowledge_base.pl) contains only domain knowledge and no reasoning code. The inference engine (engine.pl) contains only domain-independent reasoning and explanation code. Adding a new gem therefore means adding facts and rules; no engine code changes. Working memory

is declared `thread_local`, so each web request, which runs in its own thread, has a private copy and several users cannot interfere with each other.

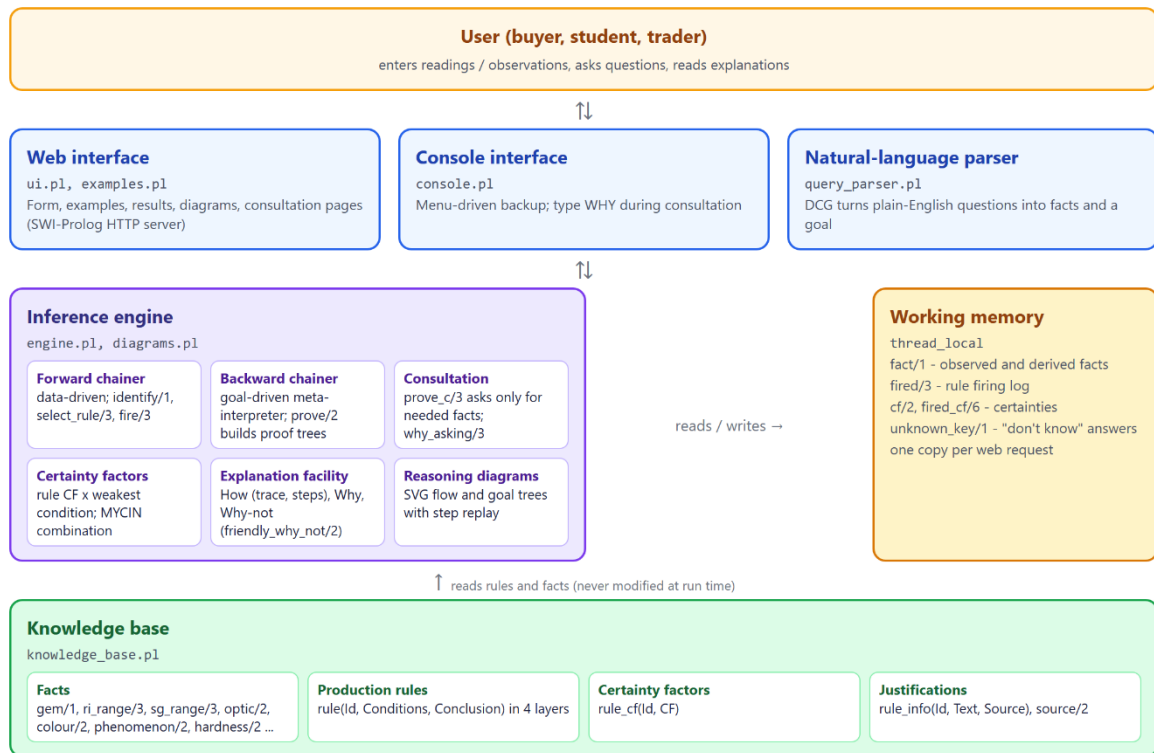


Figure 3. System architecture

Table 7. Source files

File	Responsibility
knowledge_base.pl	Facts, 62 production rules, certainty factors, justifications and sources.
engine.pl	Working memory, forward chaining, backward chaining meta-interpreter, consultation, certainty factors, How/Why/Why-not explanations.
diagrams.pl	Generates the forward-chaining flow diagram and the backward-chaining goal tree as SVG.
query_parser.pl	DCG that parses plain-English questions.
ui.pl, examples.pl	Web interface (SWI-Prolog HTTP server), example stones and gem pictures.
console.pl	Console interface.
tests.pl	47 automated tests (plunit).
main.pl	Loads all modules.

3.2 Forward chaining (Identify mode)

In Identify mode the user's observations are asserted into working memory and the engine runs the recognise-act cycle until it reaches a fixed point:

```
forward_chain :-
    (   select_rule(Id, Conds, Concl)      % recognise
->    fire(Id, Conds, Concl),              % act
        forward_chain                    % next cycle
    ;   true                              % no rule can fire: stop
    ).

select_rule(Id, Conds, Concl) :-
    rule(Id, Conds, Concl),               % next rule, in knowledge-base order
    all_true(Conds),                      % all its conditions are true
    ground(Concl),                        % conclusion fully known
    \+ fired(Id, _, Concl).               % not fired before (refractoriness)
```

Conflict resolution is by rule order: in each cycle the first applicable rule in the knowledge base fires. Because the rules are ordered by layer, all species rules fire before variety rules, which fire before origin and advice rules. This ordering is what makes the negative conditions safe: for example, rule r11 (glass) requires `not(species(_))`, and by the time it is considered every species rule has already had the chance to fire. **Refractoriness** prevents a rule from firing twice for the same conclusion; a different rule may still support an existing fact, in which case its certainty is combined into that fact. `fire/3` asserts the conclusion, records the firing in `fired/3` for the explanation facility, and stores the certainty calculation in `fired_cf/6`.

For the input RI 1.765, SG 4.00, uniaxial, blue, silk, the engine fires r1 (species corundum), r24 (blue sapphire), r46 (likely natural) and r61 (certificate advice), then stops because no further rule can add a new fact.

3.3 Backward chaining (Verify mode)

In Verify mode the user asks whether the stone is a particular gem, for example "is it a ruby?". The goal `variety(ruby)` is proved by a **meta-interpreter**, a Prolog program that interprets the rule base and builds a proof tree as it goes [12], [14]:

```

% 1. negation as failure: not(C) holds if C cannot be proved
prove(not(C), negated(C)) :- !, \+ prove(C, _).

% 2. a computed test, e.g. ri_in(corundum)
prove(C, test(C)) :- test_condition(C), !, test(C).

% 3. a fact the user entered
prove(C, given(C)) :- input_fact(C), !, fact(C).

% 4. use a rule that concludes C and prove its conditions
prove(C, by(Id, C, Proofs)) :-
    rule(Id, Conds, C),
    prove_all(Conds, Proofs).

```

Each clause handles one kind of goal, and the second argument records how it was proved:

- **Clause 1:** a negated condition such as `not(phenomenon(star))` is proved when the inner goal *cannot* be proved.
- **Clause 2:** a test such as `ri_in(corundum)` is proved by computation (comparing the reading with the stored range).
- **Clause 3:** an input such as `observed_colour(red)` is proved if the user entered it.
- **Clause 4:** any other goal is proved by finding a rule that concludes it and proving all of that rule's conditions in turn (`prove_all/2` calls `prove/2` on each one).

To prove a goal the engine selects a rule whose conclusion unifies with it and recursively proves its conditions as sub-goals, backtracking to the next rule if a condition fails. The proof tree `by(r21, variety(ruby), [by(r1, species(corundum), [...]), given(observed_colour(red)), negated(phenomenon(star))])` is the basis of the How explanation and the backward-chaining diagram. When the goal fails, `why_not/3` examines every rule that could have concluded it, finds the **first failed condition** of each, and, if that condition is itself a derived fact, explains recursively why that sub-goal failed.

3.4 Interactive consultation (Consult mode)

The third mode reproduces the MYCIN-style consultation [8]. `prove_c/3` is a variant of the meta-interpreter that keeps a **stack of the rules being tried**. When a condition needs a property the user has not yet supplied, it throws `ask(Property, Stack)`:

```

ensure_known(Stack, K) :-
    input_key(K, F),
    (   fact(F)          -> true                % already answered
;   unknown_key(K) -> true                % user said "don't know"
;   throw(ask(K, Stack))                % ask the user now
    ).

```

In words: if the property is already known, or the user has said "don't know", the proof simply continues; otherwise the proof is interrupted with a question. The interface asks the question and restarts the consultation with the new answer. Questions are therefore asked only when, and only if, a rule being tested needs them. When the goal is "find the species", backward chaining naturally performs **hypothesise-and-test**: it tries r1 (is it corundum?), asks for the optic character, and if the answer rules corundum out it abandons that hypothesis and moves on to r2 (is it spinel?) without asking anything about corundum again. The stack gives the answer to "Why are you asking this?" (Section 4.3).

3.5 Certainty factors and ranked candidates

When RI, SG or optic character is missing, no species rule can fire. Rule r14 then makes every gem that is *consistent* with all supplied properties a candidate with a low base certainty (0.20), and evidence rules r15-r19 add certainty for each supplied property that positively matches. For a red, singly refractive stone the calculation for spinel is:

```

r14 fits all inputs (rule 0.20) CF = 0.20
r15 optic matches (rule 0.30) CF = 0.20 + 0.30 x (1 - 0.20) = 0.44
r19 typical colour (rule 0.15) CF = 0.44 + 0.15 x (1 - 0.44) = 0.524

```

Garnet receives the same evidence, so the system ranks "Spinel 52%, Garnet 52%". This tie is honest: these observations genuinely cannot separate the two, and the system advises measuring RI and SG. With uniaxial and blue it ranks corundum, zircon and beryl (52%) above tourmaline (44%), because blue is not a typical tourmaline colour. An evidence rule's own conclusion is excluded from its weakest-condition calculation, since a belief is not evidence for itself.

Most useful next measurement. A ranked list is only a partial answer, so the system also tells the user what to measure next. For each core measurement the user has not given (RI, SG, optic character), `next_measurement/4` counts how many pairs of candidates it would tell apart. Two gems are told apart by optic character when their optic characters differ, and by RI or SG when

their ranges overlap by less than half of the narrower range. The measurement that separates the most pairs is recommended, together with the range each candidate would show. For the red, singly refractive stone above, the SG ranges of spinel (3.55-3.70) and garnet (3.57-4.30) overlap heavily but their RI ranges hardly do, so the system recommends measuring the refractive index (Figure 4). This mirrors how a gemmologist chooses the test that best discriminates between the remaining hypotheses.

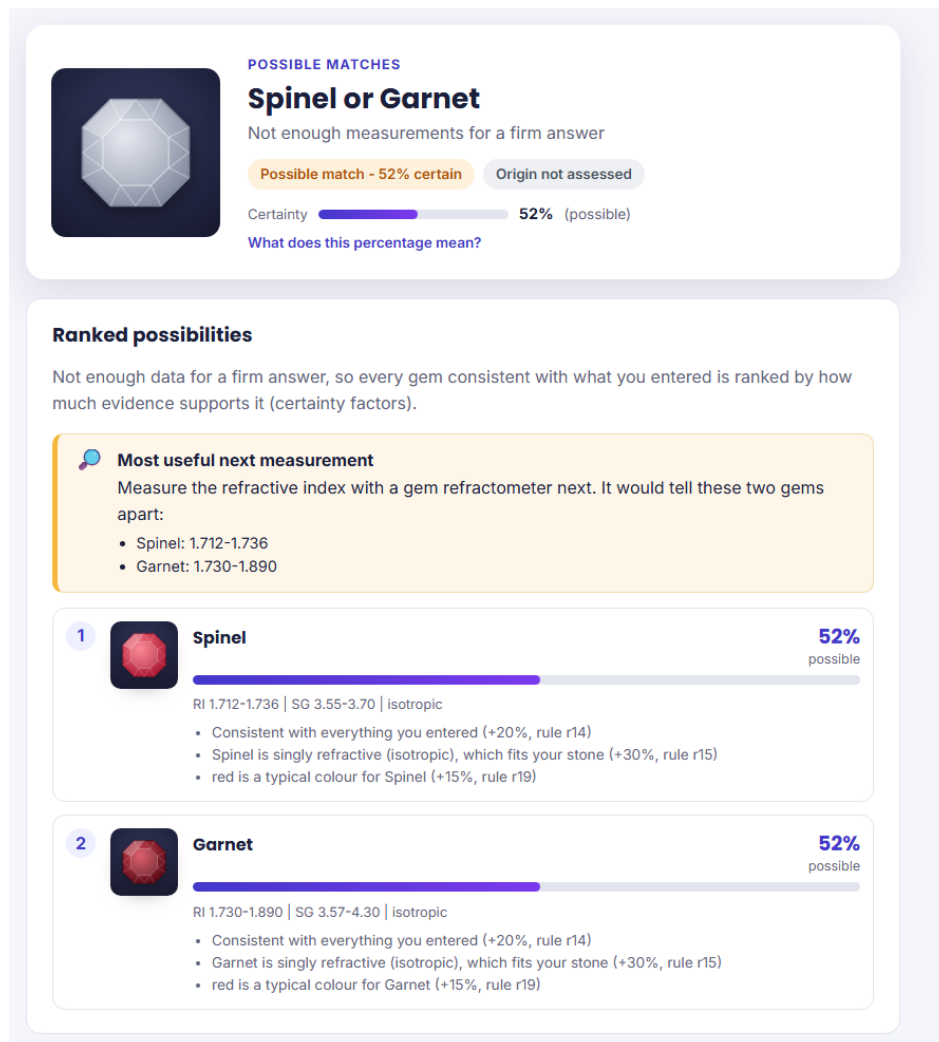


Figure 4. Ranked candidates with certainty factors, and the most useful next measurement

3.6 Natural-language interface

Users can type a question such as "Is my stone a sapphire if the RI is 1.765 and it is blue?". The question is tokenised and parsed by a Definite Clause Grammar [15] in `query_parser.pl`. The grammar recognises goal phrases ("is my stone a", "could it be"), property phrases ("RI is 1.765", "singly refractive", "curved growth lines") and gem names, and skips words it does not know, so word order is flexible:

```

items([I|Is])      --> item(I), !, items(Is).
items([skip(W)|Is]) --> [W], !, items(Is).    % unknown word: skip it
items([])          --> [].

item(goal(T))      --> goal_phrase(T).
item(fact(F))       --> fact_phrase(F).

goal_phrase(T)     --> [is], stone_ref, article, target(T).
fact_phrase(ri(N)) --> ri_word, link, number(N).

```

In DCG notation, `A --> B, C` means "an A is a B followed by a C", and words in square brackets such as `[is]` must appear literally in the sentence. So `items` reads the sentence one piece at a time: if the next words form a goal or a property phrase it keeps them, otherwise it skips one word and carries on. `goal_phrase` matches, for example, "is" + "my stone" + "a" + gem name, and `fact_phrase` matches "RI" + optional linking words ("is", "of", "about") + a number.

The example question is parsed as the goal `verify(blue_sapphire)` with facts `[ri(1.765), observed_colour(blue)]` and answered by backward chaining. Gem names are matched longest first, so "star sapphire" is not mistaken for "sapphire".

3.7 User interface

The main interface is a web application served by SWI-Prolog's built-in HTTP library, so nothing beyond SWI-Prolog needs to be installed. The home page introduces the system, explains the two chaining methods in plain terms, and provides a form in which each field explains what the property means and how to measure it (Figures 5 and 6). Result pages present the conclusion, certainty and origin as badges, the recommendation, a reasoning diagram and the plain-language reasoning steps (Figure 7). A menu-driven console interface provides the same functions as a backup. Full instructions are in the separate User Manual.

Online deployment. Because all reasoning runs on the server and the browser only receives finished pages, users do not need Prolog installed. The system is packaged as a Docker container based on the official SWI-Prolog image and deployed on the Render cloud platform at <https://gemstone-identification-assistant.onrender.com>. A `serve/0` entry point starts the server on the port supplied by the host. Working memory is thread-local, so several users can consult the system at the same time without affecting each other's results.

Several features were added specifically to make the system usable by non-experts:

- **Pictures instead of terms.** Colour is chosen from swatches, optical effects and inclusions from small drawings of what they look like under a loupe, and the optic character from drawings of what is seen in the polariscope and conoscope ("stays dark", "blinks light and dark", "dark cross", "curved bars"), because a beginner may not know the words "silk", "fingerprint" or "biaxial" (Figure 6).
- **Checking readings as they are typed.** As an RI or SG value is entered, the page immediately says which gems it fits, warns about unusual values, and blocks impossible ones once the box is left (Figure 8). Under each box a "normal range" line gives the range covered by the knowledge base, with a link to the range of every gem. The ranges are sent from the knowledge base, so the page and the rules always agree.
- **Certainty in plain words.** Every certainty factor is shown with a phrase such as "90% (very likely)" or "52% (possible)", and the scale is explained on request, including that a certainty factor is not a statistical probability.
- **An "over the limit" option** for refractometer readings (Section 2.3).
- **Guidance when data is missing:** ranked candidates and the most useful next measurement (Section 3.5).
- **Consultation progress and Back.** The step-by-step mode shows "Question 2 of at most 6" with a progress bar, and a Back button to change the previous answer.
- **Print / Save as PDF,** producing a clean, dated printout that a buyer can take to a seller or a laboratory.
- **Recent stones,** remembered only in the user's own browser, and a layout with a menu button for phones.

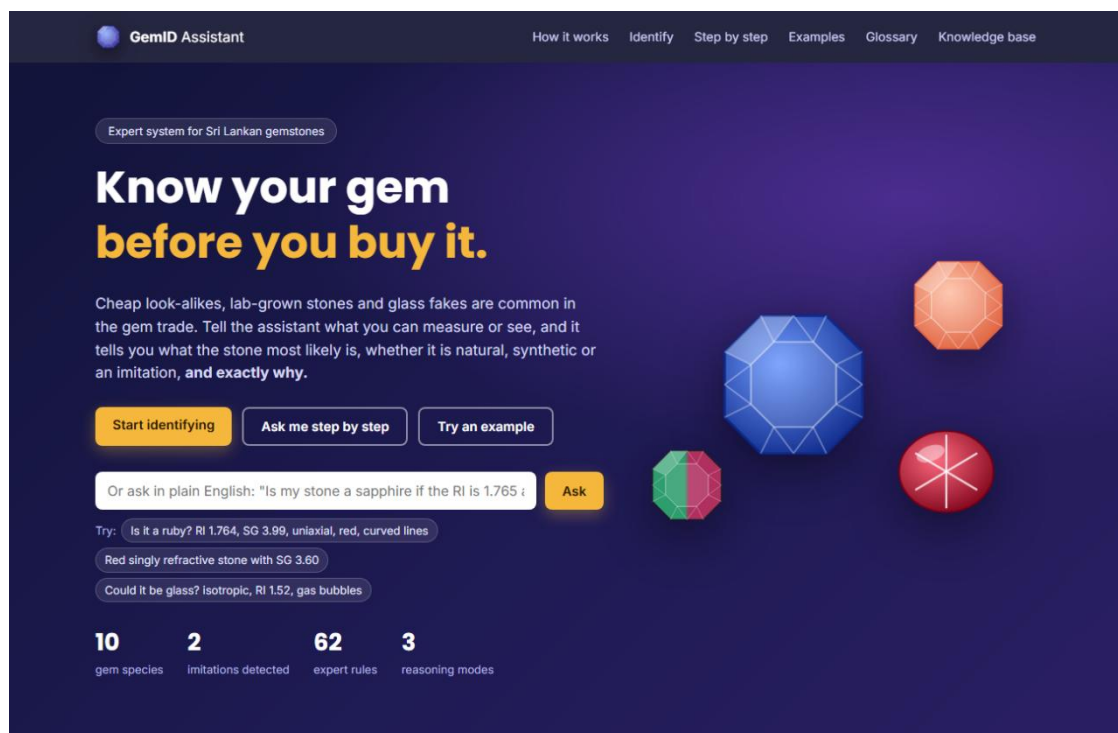


Figure 5. Home page introducing the system, with the plain-English question box

Describe your stone

Every field is optional. The more you fill in, the more confident the answer.

1 Instrument readings

Refractive index (RI)

How strongly the stone bends light. Each gem has its own range.

☐ Over the limit (no reading)

Normal range: gems 1.515 - 1.990 (with imitations 1.440 - 2.180). Most refractometers read up to about 1.81. [See the range for each gem](#)

HOW Read it from a gem refractometer (use the highest reading).

Specific gravity (SG)

How heavy the stone is compared with the same volume of water.

Normal range: gems 2.55 - 4.80 (with imitations 2.20 - 6.00).
[See the range for each gem](#)

HOW Weigh the stone in air and in water (hydrostatic balance).

2 What you can see

Optic character

Whether light passes through as one ray or is split in two. Pick the picture that matches what you see.

Not checked

Stays dark
singly
refractive

Blinks
light/dark
doubly
refractive

Dark cross
uniaxial

Curved bars
biaxial

HOW Turn the stone between crossed filters in a polariscope. A conoscope (or a glass sphere held over the stone) shows the cross or curved bars.

Colour

The main body colour of the stone. Pick the closest swatch.

Not checked

Red

Pink

Orange

Yellow

Green

Blue

Purple

Brown

Black

White

Colourless

HOW Look at it in daylight against a white background.

Optical effect

Special light effects. Pick the picture that looks like your stone.

Not checked

None

Star

Cat's-eye

Colour
change

Moonstone
glow

HOW Look under a single light source and rotate the stone.

Inclusions

Tiny features inside the stone that reveal how it formed. Pick the sketch closest to what you see.

Not checked

Clean

Silk

Crystals

Fingerprints

Gas bubbles

Curved lines

Swirl marks

Discoid
fractures

HOW Examine it with a 10x jeweller's loupe.

Identify my stone

Find the best match using every rule in the knowledge base.

Identify stone

Check a claim

Test whether the stone could be one particular gem.

Verify

Ask me step by step

The system asks only the questions it needs, and you can ask it why.

Start

[Clear the form](#)

Figure 6. Input form: instrument readings, and picture choices for optic character, colour, optical effect and inclusions

MOST LIKELY

Blue sapphire

Gem species: Corundum

Strong match - 90% certain

Likely natural (76%)

Certainty

90% (very likely)

What does this percentage mean?

- 80-100% Very likely: the evidence strongly supports it.
- 60-79% Likely: good evidence, with some doubt.
- 40-59% Possible: some evidence; take more measurements.
- 20-39% Weak evidence: treat it as a first guess only.
- below 20% Very weak: not enough evidence to go on.

It is a certainty factor: how strongly the expert rules believe the conclusion, based on the evidence you gave. It is not a statistical probability.

What you should do

- Features are consistent with a natural stone. Obtain an NGJA (National Gem and Jewellery Authority) or other accredited laboratory certificate before purchase.

What you told us

RI 1.765

SG 4.0

Optic Uniaxial

Colour Blue

Effect not given

Inclusions Silk

Figure 7. Identification result for a natural blue sapphire, with the certainty scale explained

1 Instrument readings

Refractive index (RI)

How strongly the stone bends light. Each gem has its own range.

1.72

✓ Fits: Spinel, Glass.

☐ Over the limit (no reading)

Normal range: gems 1.515 - 1.990 (with imitations 1.440 - 2.180). Most refractometers read up to about 1.81. [See the range for each gem](#)

HOW Read it from a gem refractometer (use the highest reading).

Specific gravity (SG)

How heavy the stone is compared with the same volume of water.

9

⚠ SG must be between 1.0 and 8.0.

Normal range: gems 2.55 - 4.80 (with imitations 2.20 - 6.00). [See the range for each gem](#)

HOW Weigh the stone in air and in water (hydrostatic balance).

1 Instrument readings

Refractive index (RI)

How strongly the stone bends light. Each gem has its own range.

e.g. 1.765

✓ Over the limit: fits Garnet, Zircon, Cubic zirconia.

☒ Over the limit (no reading)

Normal range: gems 1.515 - 1.990 (with imitations 1.440 - 2.180). Most refractometers read up to about 1.81. [See the range for each gem](#)

HOW Read it from a gem refractometer (use the highest reading).

Specific gravity (SG)

How heavy the stone is compared with the same volume of water.

1.9

⚠ Unusual reading: no gem in the knowledge base has an SG of 1.9. You can still continue, but please check the reading.

Normal range: gems 2.55 - 4.80 (with imitations 2.20 - 6.00). [See the range for each gem](#)

HOW Weigh the stone in air and in water (hydrostatic balance).

Figure 8. Readings checked as they are typed: an RI that fits spinel and glass, an impossible SG (top); "over the limit" ticked and an unusual SG (bottom)

3.8 Logic programming concepts used

Table 8. Where the system uses core Prolog concepts

Concept	Where and how it is used
Unification	Matching rule conclusions to goals in <code>prove/2</code> (<code>engine.pl</code> , line 375); a rule with variables, e.g. <code>r14 candidate(G)</code> , is instantiated once per matching gem.
Backtracking	<code>select_rule/3</code> (line 176) backtracks through rules and variable bindings until all conditions hold; <code>prove/2</code> backtracks to the next rule when a sub-goal fails.
Recursion	<code>forward_chain/0</code> (line 167) recurses once per cycle; <code>prove_all/2</code> , <code>why_not/3</code> (line 387) and the DCG <code>items//1</code> recurse over lists and trees.
Negation as failure	<code>test(not(C)) :- \+ holds(C)</code> (line 114) and <code>prove(not(C), ...)</code> (line 372) implement conditions such as <code>not(phenomenon(star))</code> .
Dynamic database	Working memory <code>fact/1</code> , <code>fired/3</code> , <code>cf/2</code> is changed with <code>assertz/retract</code> in <code>load_inputs/1</code> (line 98) and <code>fire/3</code> (line 183); declared <code>thread_local</code> (lines 37-41).
Meta-interpreter	<code>prove/2</code> and <code>prove_c/3</code> interpret rules stored as data and build proof trees.
Exceptions	The consultation throws <code>ask(K, Stack)</code> from inside the proof (line 293) and catches it in <code>consult/4</code> (line 264).
Definite Clause Grammar	<code>query_parser.pl</code> , <code>items//1</code> (line 58) and <code>goal_phrase//1</code> (line 66).
Higher-order predicates	<code>findall/3</code> , <code>maplist/3</code> , <code>convlist/3</code> , <code>foldl/4</code> for collecting explanations and validating input.

4. Explanation Facility

The brief requires the explanation to show the chain **user input** → **facts** → **rules fired** → **reasoning** → **conclusion**. The system provides this chain at two levels of detail, for users and for examiners, and answers the three classic explanation questions: How, Why and Why not [11], [12].

4.1 How: the reasoning trace

The technical trace is generated from the `fired/3` log. For each fired rule it lists the rule's conditions as they were satisfied, the justification and source, and the certainty calculation:

```

USER INPUT

  RI = 1.765 / SG = 4.0 / Optic character = uniaxial / Colour = blue / Inclusion = silk
FACTS (WORKING MEMORY AT START)

  ri(1.765), sg(4.0), optic(uniaxial), observed_colour(blue), inclusion(silk)
RULES FIRED (IN ORDER)

  r1 -> species(corundum)

    because: optic character is uniaxial

      RI 1.765 lies in the corundum range 1.757-1.780 (+/-0.005)

      SG 4.0 lies in the corundum range 3.95-4.10 (+/-0.03)

    why: Corundum is doubly refractive (uniaxial) with RI about 1.762-1.770 and SG
    about 4.00. [source: gia]

    certainty: rule CF 0.95 x weakest condition 1.00 = 0.95

  r24 -> variety(blue_sapphire)

    because: species(corundum) (established earlier); colour is blue;
    phenomenon(star) is not established

  r46 -> origin(likely_natural)

    because: species(corundum) (established earlier); inclusion seen: silk

  r61 -> advice(get_certificate)
CONCLUSION

  Blue sapphire (Corundum) - Strong match; Origin: Likely natural

  Recommendation: obtain an NGJA or other accredited laboratory certificate before
  purchase.

```

The same information is presented to users as numbered plain-language steps (Figure 9), for example "Its refractive index (1.765) is within the Corundum range of 1.757-1.780", each tagged with its rule, certainty and source.

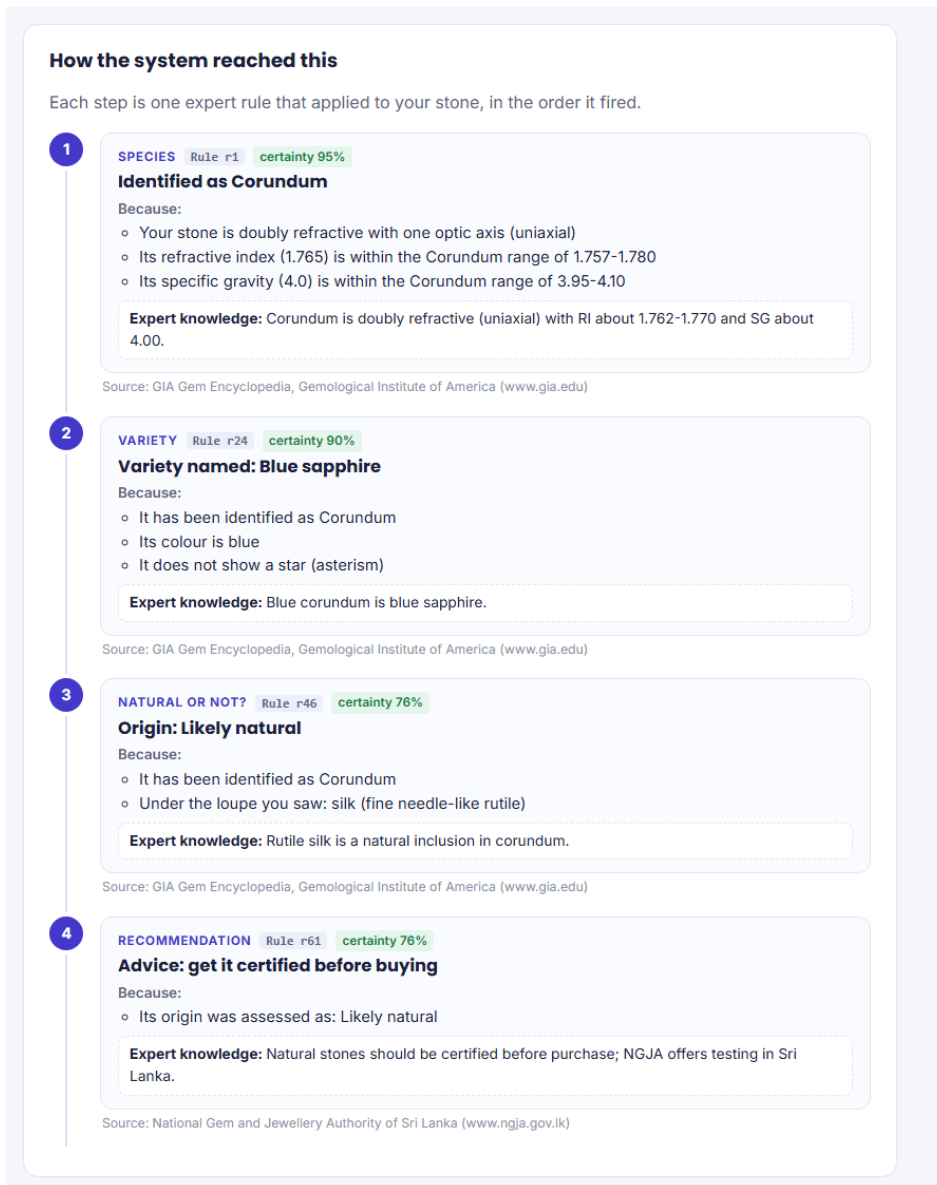


Figure 9. Plain-language How explanation

4.2 Why: explaining a question

During a consultation the user can ask why a question is being asked. The answer is built from the stack of rules being tried (Figure 10):

WHY I AM ASKING:

- I am trying to find out: is it Ruby? Rule r21 says: IF Species: Corundum AND Colour: Red
AND NOT Effect: Star THEN Variety: Ruby.
- To use that rule I first need to establish: is it Corundum? Rule r1 says: IF Optic: Uniaxial
AND RI in Corundum range AND SG in Corundum range THEN Species: Corundum.
- To test rule r1 I need to know: the optic character.

Question 1 of at most 6

QUESTION 1 testing rule r1

What is the optic character of your stone?

Optic character
Whether light passes through as one ray or is split in two. Pick the picture that matches what you see.

☐ Not checked

☐ Stays dark
singly refractive

☐ Blinks light/dark
doubly refractive

☐ Dark cross
uniaxial

☐ Curved bars
biaxial

HOW Turn the stone between crossed filters in a polariscope. A conoscope (or a glass sphere held over the stone) shows the cross or curved bars.

Answer **I don't know**

▼ **Why are you asking this?**

1. I am trying to find out: is it Ruby? Rule r21 says: IF Species: Corundum AND Colour: Red AND NOT Effect: Star THEN Variety: Ruby.
2. To use that rule I first need to establish: is it Corundum? Rule r1 says: IF Optic: Uniaxial AND RI in Corundum range AND SG in Corundum range THEN Species: Corundum.
3. To test rule r1 I need to know: the optic character.

Figure 10. Consultation question with the Why explanation

4.3 Why not: explaining a rejected alternative

When a verification fails, the system names the failed condition and traces it back through the rules. For a red spinel (isotropic, RI 1.718, SG 3.60) checked as a ruby, it answers: *"No - your stone does not match Ruby. Main reason: it must be doubly refractive with one optic axis (uniaxial), but you entered singly refractive (isotropic)"*. It then shows that rule r21 needs corundum and rule r1 failed on the optic character (Figure 11), and adds what the data points to instead (red spinel).

Why not?

The knowledge base has one way to prove a stone is Ruby. Each was checked against your data:

- It must first be identified as Corundum, which fails because:

Rule r21

X

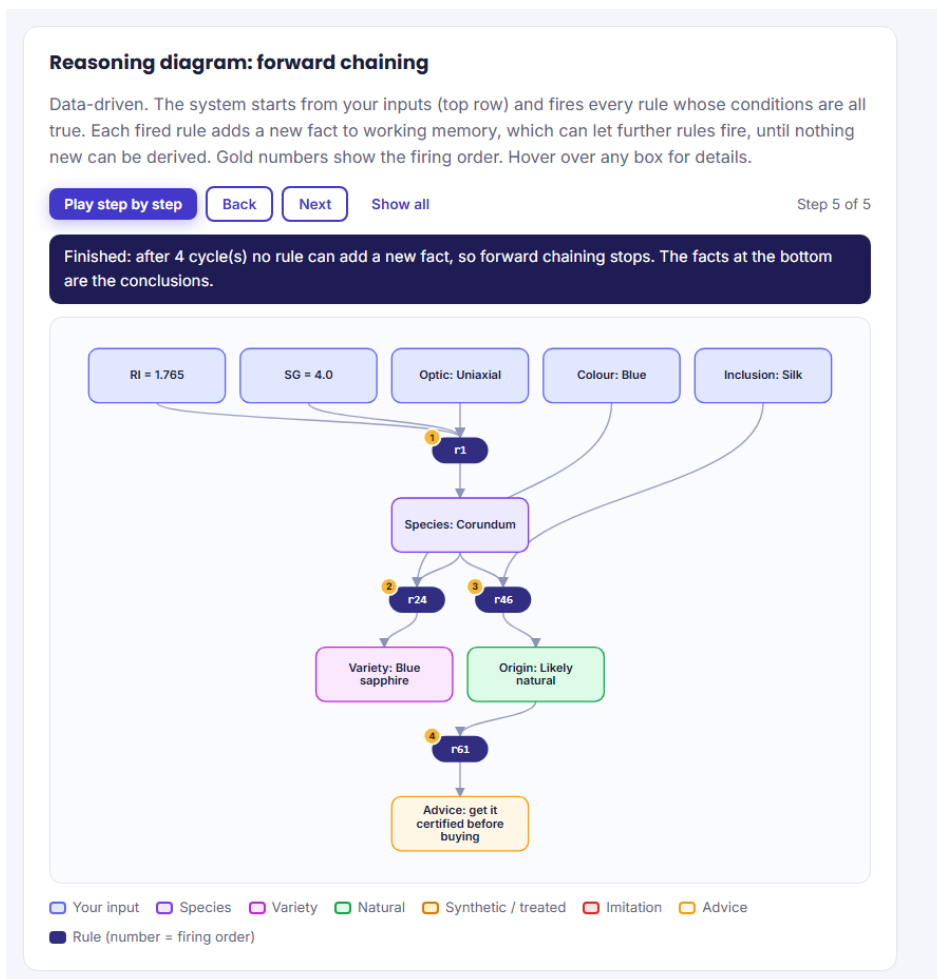
It must be doubly refractive with one optic axis (uniaxial), but you entered singly refractive (isotropic)

Rule r1

Figure 11. Why-not explanation for a failed verification

4.4 Reasoning diagrams

Each result page includes a diagram generated from the actual reasoning for the user's own input, with Play, Back and Next controls that replay it one step at a time with a caption. The forward-chaining diagram (Figure 12) shows input facts at the top, each fired rule numbered in firing order, and the facts it added. The backward-chaining diagram (Figure 13) shows the goal tree: the question, the rules tried, and each condition marked as proved, failed or never checked, which makes visible that backward chaining abandons a rule at its first failed condition.



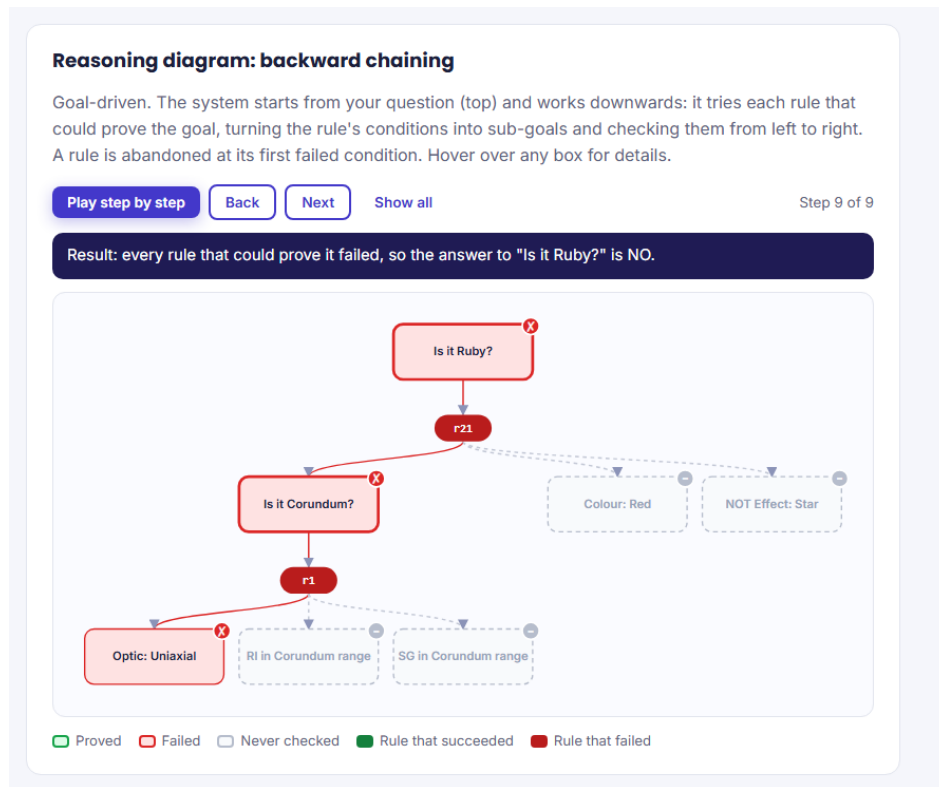


Figure 13. Backward-chaining goal tree for "is it a ruby?" with red spinel data

5. Testing and Evaluation

5.1 Test strategy

Testing was carried out at three levels: (1) documented manual test cases that exercise each type of scenario through the complete system, (2) automated unit tests with SWI-Prolog's plunit library that are re-run after every change, and (3) interface testing of every web page, including invalid input, in a browser. The 15 example stones built into the home page double as a regression set: each was run through both forward and backward chaining and produced its expected result.

5.2 Manual test cases

Table 9 lists the test cases with their expected and actual results. The actual output and fired rules were produced by running the system on each input.

Table 9. Manual test cases

TC	Scenario	Input	Expected output	Actual output	Rules fired	Result
1	Natural blue sapphire	RI 1.765, SG 4.00, uniaxial, blue, silk	Blue sapphire, likely natural, certificate advice	Blue sapphire (90%); origin: Likely natural	r1, r24, r46, r61	Pass
2	Spinel, not garnet	RI 1.718, SG 3.60, isotropic, red	Red spinel; garnet rejected	Red spinel (81%); origin: not determined	r2, r37, r52, r66	Pass
3	Garnet whose SG overlaps corundum	RI 1.79, SG 4.05, isotropic	Garnet, not corundum	Garnet (90%); origin: not determined	r3, r52, r66	Pass
4	Glass imitation	RI 1.60, isotropic, gas bubbles	Imitation: glass	Glass (85%); origin: Imitation	r11, r45, r63	Pass
5	Synthetic ruby	RI 1.765, SG 4.00, uniaxial, red, curved lines	Ruby, synthetic	Ruby (90%); origin: Synthetic	r1, r21, r41, r62	Pass
6	Star sapphire	RI 1.765, SG 4.00, uniaxial, blue, star	Star sapphire (not blue sapphire)	Star sapphire (90%)	r1, r23, r52, r66	Pass
7	Missing RI and SG	isotropic, red	Ranked candidates, measure more	Ranked: Spinel 52%, Garnet 52%	r14, r15, r19, r70	Pass
8	No match	RI 1.40, SG 1.50, uniaxial	Unknown, laboratory test advised	We could not identify this stone	r72	Pass
9	Verify - true	"Is it ruby?" with ruby data	Yes, with proof tree	YES (90%) with proof tree	r21, r1	Pass
10	Verify - false	"Is it spinel?" with sapphire data	No, names failed condition	NO - must be singly refractive (isotropic), but you entered uniaxial	r2 (failed)	Pass
11	Alexandrite	RI 1.748, SG 3.73, biaxial, green, colour change	Alexandrite, confirm-colour-change advice	Alexandrite (86%)	r4, r29, r52, r66, r68	Pass
12	Moonstone	RI 1.52, SG 2.57, biaxial, colourless, adularescence	Moonstone	Moonstone (86%)	r9, r31, r52, r66	Pass
13	Cubic zirconia	SG 5.80, isotropic, colourless	Imitation: cubic zirconia	Cubic zirconia (95%); origin: Imitation	r13, r45, r63	Pass

TC	Scenario	Input	Expected output	Actual output	Rules fired	Result
14	Heat-treated sapphire	RI 1.764, SG 4.00, uniaxial, blue, discoid fractures	Blue sapphire, natural, heated, disclosure advice	Blue sapphire (90%); Likely natural	r1, r24, r49, r50, r61, r67	Pass
15	Ambiguous spinel/garnet	RI 1.733, SG 3.62, isotropic	Both species, re-measure advice	Ranked: Spinel 90%, Garnet 90%	r2, r3, r52, r66, r71	Pass
16	Edge-of-range reading	RI 1.783 (inside tolerance only), SG 4.00, uniaxial, blue	Blue sapphire with reduced certainty	Blue sapphire (63%)	r1, r24, r52, r66	Pass
17	Consultation stops early	Goal ruby; answer optic = isotropic	No, without asking RI, SG or colour	Answer NO after 1 question; RI, SG, colour never asked	r21, r1 (failed)	Pass
18	Plain-English question	"Is my stone a sapphire if the RI is 1.765 and it is blue?"	Goal blue sapphire, facts RI and colour	Parsed as verify(blue_sapphire) with [ri(1.765), observed_colour(blue)]	-	Pass
19	RI over the refractometer limit	RI over the limit, SG 4.70, uniaxial, blue, crystals	Blue zircon, not corundum	Blue zircon (65%); origin: Likely natural	r5, r38, r47, r61	Pass
20	Next-measurement hint	isotropic, red (no RI or SG)	Spinel and garnet; advise measuring RI	Candidates spinel, garnet; suggests measuring RI next	r14, r15, r19, r70	Pass

5.3 Automated tests

The plunit suite in tests.pl contains 47 tests in five units (Table 10). All 47 pass (Appendix B). The suite is run with `run_tests`.

Table 10. Automated test suites

Unit	Tests	What is checked
identify_forward_chaining	18	Species, varieties, imitations, origin, treatment, partial data, ambiguity, no match, refractoriness, explanation generation.
verify_backward_chaining	7	Proof trees, failed goals, nested why-not reasons, plain-language explanations.
certainty_factors	11	Rule CF propagation, tolerance penalty, MYCIN combination, ranking, proof certainty, certainty words, "over the limit" RI readings, most useful next measurement.
consultation_why	6	Question order, early stopping, "don't know" answers, hypothesise-and-test, Why text.

Unit	Tests	What is checked
natural_language_dcg	5	Verify and identify questions, longest-name matching, multi-word phrases, unknown words.

5.4 Defects found and fixed

Testing found several real defects, which were fixed and, where possible, covered by a regression test (Table 11).

Table 11. Defects found during testing

Defect	Cause	Fix
Evidence rules added almost no certainty ($0.3 \times 0.2 = 0.06$)	An evidence rule's own conclusion (the candidate) was counted as its weakest condition.	Exclude the conclusion from its own weakest-condition calculation.
Consultation page returned "Internal server error"	A nested list of hidden form fields was misread by the HTML generator as attributes.	Flatten the element list; the same fix restored a missing default option in two drop-down menus.
Navigation bar and example gallery collapsed in the browser	Two CSS class names were shared by unrelated elements.	Rename the conflicting classes.
Console menu looped forever when input ended	End of input was treated as menu choice 1.	Treat end of input as "Quit" in menus.
Diagram coordinates were written as unevaluated expressions	Arithmetic was passed to format/2 without evaluation.	Evaluate with is/2 before formatting.
Pages were wider than a phone screen, hiding the menu	Long example-question chips could not wrap.	Allow wrapping; add a menu button for small screens.
A stray "Fits: ..." message appeared on consultation questions	The live reading check attached itself to a hidden field carrying an earlier answer.	Check only visible input fields.

5.5 Evaluation against the objectives

Table 12. Evaluation against the objectives

Objective	Status	Evidence
O1 Knowledge acquisition documented	Met	Table 3; rule_info/3 stores the source of every rule; Appendix A.
O2 At least 20 facts and 20 rules	Exceeded	297 facts, 62 rules in four chained layers.

Objective	Status	Evidence
O3 Forward and backward chaining	Met	Identify, Verify and Consult modes over one rule base; TC1-17.
O4 Explanation facility	Met	How, Why and Why-not at technical and plain-language levels; diagrams.
O5 Incomplete data and confidence	Met	Certainty factors, ranked candidates, next-measurement hint, "over the limit" readings; TC7, 15, 16, 19, 20.
O6 User-friendly interface	Met	Web and console interfaces, plain-English questions, example stones, picture choices, live input checks, print/save, phone layout.
O7 Testing and user manual	Met	20 manual and 47 automated tests; separate User Manual.

The main gap in the evaluation is that all tests use textbook values. The system has not yet been validated against independently certified stones or laboratory reports. Such a validation study, reporting an accuracy table with analysis of any failures, is the most important next step (Section 7.4).

6. User Manual

The complete user manual is included in this report as **Appendix D**. It describes the two ways to use the system: the online version at **<https://gemstone-identification-assistant.onrender.com>**, which needs only a web browser, and running it on one's own computer by downloading the project ZIP file from GitHub and starting it from the Command Prompt. It covers the required software, installation of SWI-Prolog, starting the web and console interfaces, the meaning of each input field and where to obtain it, worked examples with screenshots, the three reasoning modes, troubleshooting, and how to run the automated tests.

7. Discussion

7.1 Modelling expert cognition

The three reasoning modes correspond to recognisable patterns of expert thinking. When a gemmologist first receives an unknown stone, they gather observations and let the evidence suggest conclusions. This is data-driven reasoning, modelled by forward chaining. Once a hypothesis forms ("this could be a sapphire"), expert reasoning becomes **hypothesise-and-test**: the expert selects the tests that would confirm or rule out that hypothesis and abandons it as

soon as a test contradicts it. Studies of expert diagnostic reasoning describe the same hypothetico-deductive pattern [18]. The consultation mode models it directly. It forms a hypothesis from the rule order, asks only for the evidence that hypothesis needs, and moves to the next hypothesis on failure. Test case 17 shows the effect: an isotropic reading rules out ruby after a single question.

Human experts also reason with uncertain evidence and weigh several weak clues. The certainty factors model this: no single colour observation identifies a gem, but several weak matches together raise one candidate above another. Finally, experts can justify their conclusions. The explanation facility gives the system the same ability to justify, and its explanations cite the source of each piece of knowledge.

7.2 Symbolic AI compared with machine learning

An alternative would be to train a machine-learning model, for example a neural network that classifies gem photographs [17]. Such models can be very accurate on images and can learn patterns no expert has written down. They are, however, largely "black boxes": they cannot say *why* a stone was classified as a sapphire, they need large labelled data sets that do not exist for Sri Lankan stones and their treatments, and a photograph cannot capture RI, SG or optic character. This rule-based system is transparent. Every conclusion can be traced to named rules and published sources, and it is easy to correct when an expert disagrees. Rudin [19] argues that for high-stakes decisions, interpretable models should be preferred to explaining black boxes after the fact. A buyer deciding whether to spend a large sum on a stone is making exactly that kind of decision, which is why explainability was treated as a core requirement. The weakness of the symbolic approach is the knowledge acquisition bottleneck: every rule must be written and justified by hand, and the system knows nothing outside its rules. A hybrid, in which image recognition proposes candidates that the rule base then checks and explains, is a promising direction.

7.3 Limitations

- **No validation on real certified stones.** Accuracy on real-world readings, with their measurement noise, is unknown.
- **Over-the-limit readings are coarse.** An RI "over the limit" only says the RI is above about 1.81, so it cannot separate zircon from cubic zirconia or high-RI garnets on its own; SG and optic character must do that.

- **Unused properties.** Birefringence is stored but not used as an input; pleochroism, fluorescence and absorption spectra, which gemmologists use to separate overlapping species, are not modelled.
- **Treatments.** Only heat treatment is indicated, and only from one visible feature. Glass-filled ruby, a significant fraud, is not covered.
- **One inclusion per stone.** The user can report only the dominant inclusion type.
- **Subjective certainty factors.** The CF values are the knowledge engineer's judgement and have not been calibrated against data. MYCIN-style certainty factors also lack a full probabilistic foundation [17].
- **Rule-order dependence.** Rules with negative conditions rely on the layer order for correct conflict resolution. This is documented but would need care if rules were added carelessly.
- **Consultation diagram.** For "find the species", the diagram shows the successful proof but not the hypotheses rejected before it, although the question sequence and the Why explanations do show them.

7.4 Future work

- Validate on 15-20 certified stones or laboratory reports and report accuracy with an error analysis.
- Add birefringence as an input, and rules for pleochroism, fluorescence and glass-filled ruby.
- Allow several inclusion types per stone, combining their evidence with certainty factors.
- Add a knowledge base consistency checker (conflicting, redundant or unreachable rules) and a knowledge-editing page for experts.
- Explore a hybrid system in which image recognition proposes candidates that the rule base verifies and explains.

8. Conclusion

This project designed and implemented a rule-based expert system that identifies common Sri Lankan gemstones, detects synthetics and imitations, and explains its reasoning. Knowledge from recognised gemmological sources was systematically acquired, documented and

represented as 297 facts and 62 production rules in four chained layers, with a certainty factor and a literature source for every rule. One rule base supports three reasoning modes: forward chaining for identification, backward chaining for verifying a claim, and an interactive consultation that asks only for the facts it needs. A full How, Why and Why-not explanation facility, reasoning diagrams, a plain-English question interface and a web interface make the reasoning transparent to non-experts, and features such as picture-based inputs, live checking of readings and the next-measurement hint make it practical for people without gemmological training. All 20 documented test cases and 47 automated tests pass. The system meets its objectives as a tool for a first opinion. Its main limitations, the absence of real-world validation and several unmodelled gemmological properties, define clear next steps. Above all, the project shows why a transparent, explainable system is appropriate when a person's money depends on its answer.

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Appendix A. Complete Rule Base

Generated from knowledge_base.pl. Each row reads: IF all the conditions (joined by AND) are true, THEN the conclusion is added, with the certainty factor (CF) shown. Source keys: gia [1], webster [2], read [3], matlins [4], igs [5], gemdat [6], ngja [7], design = knowledge engineer decision. Capital letters such as G are variables: they stand for any gem, so a rule like r14 applies to every gem that satisfies it. Table A1 explains the condition terms.

Table A1. Key to the condition terms used in the rules

Term	Meaning
optic(T)	The stone's optic character is T (isotropic, uniaxial or biaxial). "Doubly refractive" fits uniaxial or biaxial.
ri_in(G), sg_in(G)	The RI (or SG) reading lies within gem G's range, allowing ± 0.005 for RI and ± 0.03 for SG. An RI "over the limit" fits gems whose range goes above about 1.81.
observed_colour(C), phenomenon(P), inclusion(I)	The user reported this colour, optical effect or inclusion.
species(G), variety(V), imitation(M)	Conclusions of earlier rules: the stone has been identified as species G, variety V or imitation material M.
origin(O), treatment(T), candidate(G)	Conclusions of earlier rules: likely origin, probable treatment, or a possible gem when data is incomplete.
not(C)	C has not been established (negation as failure).
missing(K)	The user did not supply property K (e.g. ri, sg, optic, inclusion).
partial_data	At least one of RI, SG or optic character is missing, and at least one property was supplied.
gem_type(G)	G is one of the ten gem species in the knowledge base.
optic_ok(G), ri_ok(G), sg_ok(G), colour_ok(G), phenomenon_ok(G)	The supplied optic character, RI, SG, colour or effect is consistent with gem G (true when that property was not supplied).
typical_colour_ok(G)	The reported colour is one of G's most common colours in the trade.
ambiguous_species	More than one species matched the measurements.
advice(K)	A recommendation is given; K names its wording (e.g. get_certificate).

Table A2. Complete rule base

Id	IF (all conditions true)	THEN (conclusion)	CF	Justification [source]
r1	optic(uniaxial) AND ri_in(corundum) AND sg_in(corundum)	species(corundum)	0.95	Corundum is doubly refractive (uniaxial) with RI about 1.762-1.770 and SG about 4.00. [gia]
r2	optic(isotropic) AND ri_in(spinel) AND sg_in(spinel)	species(spinel)	0.90	Spinel is singly refractive (isotropic) with RI about 1.718 and SG about 3.60. [webster]
r3	optic(isotropic) AND ri_in(garnet) AND sg_in(garnet)	species(garnet)	0.90	Garnets are singly refractive with RI 1.73-1.89 and SG 3.6-4.3 depending on the type. [webster]
r4	optic(biaxial) AND ri_in(chrysoberyl) AND sg_in(chrysoberyl)	species(chrysoberyl)	0.95	Chrysoberyl is biaxial with RI about 1.746-1.755 and SG about 3.73. [gia]
r5	optic(uniaxial) AND ri_in(zircon) AND sg_in(zircon)	species(zircon)	0.90	High zircon is uniaxial with very high RI (often over the refractometer limit) and SG about 4.7. [gemdat]
r6	optic(biaxial) AND ri_in(topaz) AND sg_in(topaz)	species(topaz)	0.95	Topaz is biaxial with RI about 1.61-1.64 and SG about 3.53. [gia]
r7	optic(uniaxial) AND ri_in(tourmaline) AND sg_in(tourmaline)	species(tourmaline)	0.90	Tourmaline is uniaxial with RI about 1.62-1.64 and SG about 3.06. [gia]
r8	optic(uniaxial) AND ri_in(quartz) AND sg_in(quartz)	species(quartz)	0.90	Quartz is uniaxial with RI 1.544-1.553 and SG 2.65. [read]
r9	optic(biaxial) AND ri_in(feldspar) AND sg_in(feldspar)	species(feldspar)	0.90	Orthoclase feldspar (moonstone) is biaxial with RI about 1.52 and SG about 2.57. [read]
r10	optic(uniaxial) AND ri_in(beryl) AND sg_in(beryl)	species(beryl)	0.90	Beryl is uniaxial with RI about 1.57-1.60 and SG about 2.7. [gemdat]
r11	optic(isotropic) AND ri_in(glass) AND inclusion(gas_bubbles) AND not(species(G))	imitation(glass)	0.85	A singly refractive stone with a glass RI and round gas bubbles, matching no gem species, is glass. [igs]

Id	IF (all conditions true)	THEN (conclusion)	CF	Justification [source]
r12	inclusion(swirl_marks) AND not(species(G))	imitation(glass)	0.80	Swirl marks (flow lines) are typical of moulded glass. [matlins]
r13	optic(isotropic) AND sg_in(cubic_zirconia)	imitation(cubic_zirconia)	0.95	Cubic zirconia is isotropic with an SG of 5.6-6.0, far heavier than any natural gem in scope. [gia]
r14	partial_data AND not(species(G)) AND not(imitation(H)) AND gem_type(I) AND optic_ok(I) AND ri_ok(I) AND sg_ok(I) AND colour_ok(I) AND phenomenon_ok(I)	candidate(I)	0.20	When RI, SG or optic character is missing, every gem consistent with ALL the supplied properties remains a candidate. [design]
r15	candidate(G) AND not(missing(optic)) AND optic_ok(G)	candidate(G)	0.30	A matching optic character is moderate evidence for a candidate. [design]
r16	candidate(G) AND not(missing(ri)) AND ri_ok(G)	candidate(G)	0.50	A matching refractive index is the strongest single test in gemmology. [read]
r17	candidate(G) AND not(missing(sg)) AND sg_ok(G)	candidate(G)	0.40	A matching specific gravity is good supporting evidence. [read]
r18	candidate(G) AND not(missing(phenomenon)) AND not(phenomenon(none)) AND phenomenon_ok(G)	candidate(G)	0.40	An optical effect the species is known for is good evidence. [gia]
r19	candidate(G) AND typical_colour_ok(G)	candidate(G)	0.15	A colour that is typical for the species is weak supporting evidence. [matlins]
r21	species(corundum) AND observed_colour(red) AND not(phenomenon(star))	variety(ruby)	0.95	Red corundum is called ruby. [gia]
r22	species(corundum) AND observed_colour(red) AND phenomenon(star)	variety(star_ruby)	0.95	Red corundum showing asterism (a star) is star ruby. [gia]
r23	species(corundum) AND phenomenon(star) AND not(observed_colour(red))	variety(star_sapphire)	0.95	Non-red corundum showing asterism is star sapphire. [gia]

Id	IF (all conditions true)	THEN (conclusion)	CF	Justification [source]
r24	species(corundum) AND observed_colour(blue) AND not(phenomenon(star))	variety(blue_sapphire)	0.95	Blue corundum is blue sapphire. [gia]
r25	species(corundum) AND observed_colour(pink) AND not(phenomenon(star))	variety(pink_sapphire)	0.85	Pink corundum (lighter than ruby) is sold as pink sapphire. [gia]
r26	species(corundum) AND observed_colour(orange) AND not(phenomenon(star))	variety(padparadscha)	0.70	Pinkish-orange to orange corundum from Sri Lanka is traded as padparadscha. [gia]
r27	species(corundum) AND observed_colour(yellow) AND not(phenomenon(star))	variety(yellow_sapphire)	0.90	Yellow corundum is yellow sapphire. [gia]
r28	species(corundum) AND observed_colour(colourless) AND not(phenomenon(star))	variety(white_sapphire)	0.90	Colourless corundum is white sapphire. [gia]
r29	species(chrysoberyl) AND phenomenon(colour_change)	variety(alexandrite)	0.90	Chrysoberyl that changes colour between daylight and incandescent light is alexandrite. [gia]
r30	species(chrysoberyl) AND phenomenon(cats_eye)	variety(cats_eye_chrysoberyl)	0.95	Chrysoberyl showing chatoyancy is cat's-eye chrysoberyl (cymophane). [gia]
r31	species(feldspar) AND phenomenon(adularescence)	variety(moonstone)	0.95	Orthoclase feldspar showing adularescence is moonstone. [gia]
r32	species(quartz) AND observed_colour(purple)	variety(amethyst)	0.95	Purple quartz is amethyst. [gia]
r33	species(quartz) AND observed_colour(yellow)	variety(citrine)	0.90	Yellow quartz is citrine. [gia]
r34	species(garnet) AND observed_colour(orange)	variety(hessonite)	0.85	Orange to brownish-orange garnet is hessonite (grossular). [gia]
r35	species(garnet) AND observed_colour(red)	variety(pyrope_almandine)	0.80	Red garnet is usually of the pyrope-almandine series. [webster]
r36	species(beryl) AND observed_colour(blue)	variety(aquamarine)	0.90	Blue beryl is aquamarine. [gia]

Id	IF (all conditions true)	THEN (conclusion)	CF	Justification [source]
r37	species(spinel) AND observed_colour(red)	variety(red_spinel)	0.90	Red spinel is a valued variety often confused with ruby. [gia]
r38	species(zircon) AND observed_colour(blue)	variety(blue_zircon)	0.85	Blue zircon is almost always produced by heating brown zircon. [gia]
r41	species(corundum) AND inclusion(curved_lines)	origin(synthetic)	0.95	Curved growth lines (striae) are diagnostic of flame-fusion (Verneuil) synthetic corundum; natural corundum grows in straight lines. [gia]
r42	species(corundum) AND inclusion(gas_bubbles)	origin(synthetic)	0.90	Round gas bubbles in corundum indicate a flame-fusion synthetic. [gia]
r43	species(spinel) AND inclusion(gas_bubbles)	origin(synthetic)	0.85	Gas bubbles in spinel indicate flame-fusion synthetic spinel. [webster]
r44	inclusion(gas_bubbles) AND not(species(G)) AND not(imitation(H))	origin(glass_or_synthetic)	0.70	Round gas bubbles in an unidentified stone indicate glass or a synthetic. [igs]
r45	imitation(G)	origin(imitation)	1.00	An identified imitation material is not a natural gemstone. [design]
r46	species(corundum) AND inclusion(silk)	origin(likely_natural)	0.80	Rutile silk is a natural inclusion in corundum. [gia]
r47	species(G) AND inclusion(crystals)	origin(likely_natural)	0.70	Included mineral crystals are a typical sign of natural growth. [matlins]
r48	species(G) AND inclusion(fingerprints)	origin(likely_natural)	0.60	Fingerprints (healed fractures) are usually natural, although flux synthetics can show similar features. [matlins]
r49	species(corundum) AND inclusion(discooid_fractures)	treatment(likely_heated)	0.80	Discooid fractures around crystals form when corundum is heated. [gia]

Id	IF (all conditions true)	THEN (conclusion)	CF	Justification [source]
r50	treatment(likely_heated)	origin(likely_natural)	0.90	Heat treatment is applied to natural stones, so a heated stone is natural (but treated). [gia]
r51	species(corundum) AND inclusion(none)	origin(needs_lab_test)	0.60	Natural corundum is rarely inclusion-free; a clean stone needs a lab test to rule out a synthetic. [igs]
r52	species(G) AND missing(inclusion)	origin(undetermined)	1.00	Without inclusion evidence the origin cannot be decided. [design]
r53	species(G) AND inclusion(none) AND not(species(corundum))	origin(undetermined)	1.00	A clean stone of this species gives no origin evidence at loupe level. [design]
r61	origin(likely_natural)	advice(get_certificate)	1.00	Natural stones should be certified before purchase; NGJA offers testing in Sri Lanka. [ngja]
r62	origin(synthetic)	advice(synthetic_price)	1.00	Synthetic gems are far cheaper than natural ones and must be sold as synthetic. [igs]
r63	origin(imitation)	advice(imitation_value)	1.00	Imitations have little gem value. [igs]
r64	origin(glass_or_synthetic)	advice(lab_test_bubbles)	1.00	Stones showing gas bubbles need laboratory confirmation. [igs]
r65	origin(needs_lab_test)	advice(lab_test_clean)	1.00	Clean corundum requires advanced laboratory tests to prove natural origin. [gia]
r66	origin(undetermined)	advice(check_inclusions)	1.00	Inclusions seen with a 10x loupe are the main field test for origin. [matlins]
r67	treatment(likely_heated)	advice(disclose_heat)	1.00	Treatments must be disclosed to the buyer. [ngja]
r68	variety(alexandrite)	advice(check_colour_change)	1.00	Alexandrite is identified by its daylight/incandescent colour change. [gia]

Id	IF (all conditions true)	THEN (conclusion)	CF	Justification [source]
r69	variety(padparadscha)	advice(padparadscha_cert)	1.00	The padparadscha name is reserved for a specific colour range. [gia]
r70	candidate(G)	advice(measure_more)	1.00	Additional measurements are needed to narrow down the candidates. [design]
r71	ambiguous_species	advice(re_measure)	1.00	Overlapping readings must be resolved by more precise measurement. [design]
r72	not(species(G)) AND not(candidate(H)) AND not(imitation(I))	advice(no_match)	1.00	When no rule identifies the stone, the system must not guess. [design]

Appendix B. Automated Test Output

```
% Start unit: identify_forward_chaining
% [1/47] identify_forward_..tural_blue_sapphire ..... passed (0.008 sec)
% [2/47] identify_forward_..2_spinel_not_garnet ..... passed (0.000 sec)
% [3/47] identify_forward_..garnet_not_corundum ..... passed (0.000 sec)
% [4/47] identify_forward_..c04_glass_imitation ..... passed (0.000 sec)
% [5/47] identify_forward_..tc05_synthetic_ruby ..... passed (0.000 sec)
% [6/47] identify_forward_..:tc06_star_sapphire ..... passed (0.000 sec)
% [7/47] identify_forward_..ri_gives_candidates ..... passed (0.000 sec)
% [8/47] identify_forward_..ining:tc08_no_match ..... passed (0.000 sec)
% [9/47] identify_forward_..ng:tc11_alexandrite ..... passed (0.000 sec)
% [10/47] identify_forward_..ning:tc12_moonstone .... passed (0.000 sec)
% [11/47] identify_forward_..tc13_cubic_zirconia .... passed (0.000 sec)
% [12/47] identify_forward_..c14_heated_sapphire .... passed (0.000 sec)
% [13/47] identify_forward_..active_unknown_axis .... passed (0.000 sec)
% [14/47] identify_forward_..guous_spinel_garnet .... passed (0.000 sec)
% [15/47] identify_forward_.._corundum_needs_lab .... passed (0.000 sec)
% [16/47] identify_forward_..etic_spinel_bubbles .... passed (0.000 sec)
% [17/47] identify_forward_..for_same_conclusion .... passed (0.000 sec)
% [18/47] identify_forward_..nation_is_generated .... passed (0.002 sec)
% End unit identify_forward_chaining: passed (0.016 sec CPU)
% Start unit: verify_backward_chaining
% [19/47] verify_backward_c..09_verify_ruby_true .... passed (0.000 sec)
% [20/47] verify_backward_c..es_failed_condition .... passed (0.000 sec)
% [21/47] verify_backward_c..false_nested_reason .... passed (0.000 sec)
% [22/47] verify_backward_c..g:tc22_verify_glass .... passed (0.000 sec)
% [23/47] verify_backward_c..fy_explanation_text .... passed (0.000 sec)
% [24/47] verify_backward_c..teps_plain_language .... passed (0.000 sec)
% [25/47] verify_backward_c..why_not_main_reason .... passed (0.000 sec)
% End unit verify_backward_chaining: passed (0.000 sec CPU)
% Start unit: certainty_factors
% [26/47] certainty_factors..pecies_cf_from_rule .... passed (0.000 sec)
% [27/47] certainty_factors..e_reading_lowers_cf .... passed (0.000 sec)
% [28/47] certainty_factors..ombines_mycin_style .... passed (0.000 sec)
% [29/47] certainty_factors_..most_certain_first .... passed (0.001 sec)
% [30/47] certainty_factors:tc30_proof_certainty .... passed (0.000 sec)
% [31/47] certainty_factors_..er_the_limit_zircon .... passed (0.000 sec)
% [32/47] certainty_factors_..les_out_low_ri_gems .... passed (0.000 sec)
```

```

% [33/47] certainty_factors..arse_over_the_limit .... passed (0.000 sec)
% [34/47] certainty_factors..ement_spinel_garnet .... passed (0.000 sec)
% [35/47] certainty_factors..t_when_it_separates .... passed (0.000 sec)
% [36/47] certainty_factors:tc42_certainty_words .... passed (0.000 sec)
% End unit certainty_factors: passed (0.000 sec CPU)
% Start unit: consultation_why
% [37/47] consultation_why:..ptic_first_for_ruby .... passed (0.000 sec)
% [38/47] consultation_why:..en_hypothesis_fails .... passed (0.000 sec)
% [39/47] consultation_why:..ks_next_needed_fact .... passed (0.000 sec)
% [40/47] consultation_why:..se_and_test_species .... passed (0.000 sec)
% [41/47] consultation_why:..t_know_is_respected .... passed (0.000 sec)
% [42/47] consultation_why:..anation_names_rules .... passed (0.000 sec)
% End unit consultation_why: passed (0.000 sec CPU)
% Start unit: natural_language_dcg
% [43/47] natural_language_..c37_verify_question .... passed (0.000 sec)
% [44/47] natural_language_..dentify_description .... passed (0.000 sec)
% [45/47] natural_language_..ngest_gem_name_wins .... passed (0.000 sec)
% [46/47] natural_language_..0_multiword_phrases .... passed (0.000 sec)
% [47/47] natural_language_..nown_words_reported .... passed (0.000 sec)
% End unit natural_language_dcg: passed (0.000 sec CPU)
% All 47 tests passed in 0.041 seconds (0.031 cpu)

```

Appendix C. Console Consultation Transcript

Consult mode, goal "find the species". The user types *why* at the first question, then answers isotropic, RI 1.718 and SG 3.60.

What is the optic character?

- 0. Don't know
 - 1. Isotropic
 - 2. Uniaxial
 - 3. Biaxial
 - 4. Doubly refractive
- (type a number, or WHY)

> WHY I AM ASKING:

- I am trying to find out: is it Corundum? Rule r1 says: IF Optic: Uniaxial AND RI in Corundum range AND SG in Corundum range THEN Species: Corundum.
- To test rule r1 I need to know: the optic character.

What is the optic character?

- 0. Don't know
 - 1. Isotropic
 - 2. Uniaxial
 - 3. Biaxial
 - 4. Doubly refractive
- (type a number, or WHY)

>

What is the refractive index (RI)? (type OTL if the refractometer shows no reading)

(type a number, press Enter if you don't know, or type WHY)

>

What is the specific gravity (SG)?

(type a number, press Enter if you don't know, or type WHY)

>

Backward chaining asked 3 of the 6 possible questions.

YES - your stone is consistent with Spinel (certainty 90% - very likely).

Every condition the expert rules require is met.

HOW IT WAS CHECKED

1. [Species] Identified as Spinel (rule r2; certainty 90%)

- Your stone is singly refractive (isotropic)
- Its refractive index (1.718) is within the Spinel range of 1.712-1.736
- Its specific gravity (3.6) is within the Spinel range of 3.55-3.70

Expert knowledge: Spinel is singly refractive (isotropic) with RI about 1.718 and SG about 3.60.

Appendix D. User Manual

This appendix is the complete user manual for the Gemstone Identification Assistant. It explains how to use the online version and how to install and run the system on your own computer, describes every part of the interface, and gives worked examples and test readings for every gem.

Online version: <https://gemstone-identification-assistant.onrender.com>

Source code: <https://github.com/Suweka/Gemstone-Identification-Assistant>

D.1 About the system

The Gemstone Identification Assistant is an expert system that helps buyers, students and small traders identify common Sri Lankan gemstones. You describe a stone (its measurements and what you can see) and the system tells you:

- what the stone most likely is (species and variety, for example *corundum - blue sapphire*);
- whether it is likely natural, synthetic (laboratory-grown) or an imitation such as glass;
- how certain it is, as a percentage;
- what you should do next, for example obtain a laboratory certificate before buying;
- exactly how it reached its answer.

It knows ten gem species (corundum, spinel, garnet, chrysoberyl, zircon, topaz, tourmaline, quartz, feldspar/moonstone and beryl), their main varieties, and two imitation materials (glass and cubic zirconia).

Important: the system gives a first opinion only. It does not replace a certificate from the National Gem and Jewellery Authority (NGJA) or another accredited gem laboratory. Always obtain a certificate before a significant purchase.

D.1.1 Three ways to use it

Mode	Use it when...	What it does
Identify	"I don't know what this stone is."	Takes everything you enter and works forwards to the most likely answer (forward chaining).

Mode	Use it when...	What it does
Verify	"The seller says it's a ruby. Is it?"	Starts from the claim and checks every condition it needs; if the claim fails it tells you exactly why (backward chaining).
Step by step	"I want to be guided."	Asks you one question at a time, only when needed, and explains why it is asking.

You can also type a question in plain English, for example *"Is my stone a sapphire if the RI is 1.765 and it is blue?"*

D.2 Two ways to use the system

The system can be used in two ways. Both give exactly the same results.

Table D1. Ways to use the system

	A. Online version (Section D.3)	B. On your own computer (Section D.4)
Installation	None. Open a link in a web browser.	Install SWI-Prolog and download the project files.
Internet	Needed.	Only for the downloads; afterwards it works offline.
Features	Web interface: Identify, Verify, Step by step, plain-English questions, examples, knowledge base.	Everything in A, plus the console interface and the automated tests.
Best for	Trying the system quickly, on any computer, tablet or phone.	Working offline, examining the code, running the tests.

D.3 Option A: use the online version (no installation)

The system is hosted on the internet, so nothing needs to be installed. Only a web browser is needed (Microsoft Edge, Google Chrome, Firefox or Safari, on a computer, tablet or phone).

1. Open your web browser.
2. Go to **<https://gemstone-identification-assistant.onrender.com>**
3. The home page appears (Figure D1). Continue from Section D.5.

First visit may be slow. The online version runs on a free hosting service that "sleeps" when nobody has used it for about 15 minutes. The first page can then take up to about a minute to appear while it wakes up; after that it is fast. If the page does not load, wait a moment and refresh.

The online version has every feature of the web interface. The console interface (Section D.16) and the automated tests (Section D.17) are only available when the system runs on your own computer (Option B). Your "recent stones" list is saved only in your own browser (Section D.13).

D.4 Option B: run the system on your own computer

D.4.1 What you need

Table D2. Requirements for running the system on your computer

Item	Requirement
Computer	Windows 10 or 11, macOS or Linux, with about 100 MB of free disk space.
SWI-Prolog	Version 9 or later (free). The system was developed and tested with SWI-Prolog 10.1.16 (64-bit).
Web browser	Any modern browser: Microsoft Edge, Google Chrome, Firefox or Safari.
Project files	The project ZIP file from GitHub (Section D.4.3).
Internet	Only for downloading SWI-Prolog and the project. The system itself runs offline; without internet the page simply uses standard fonts.
Gem instruments (optional)	Refractometer, hydrostatic balance, polariscope and a 10x loupe give the best results, but the system also works with only what you can see.

D.4.2 Install SWI-Prolog

1. Open <https://www.swi-prolog.org/download/stable> in your browser.
2. Download the installer for your computer. On Windows choose the **64-bit** installer (swipl-...-1.x64.exe).
3. Run the installer and accept the default settings.
4. **Windows only:** on the "Install Options" page select "**Add swipl to the system PATH for all users**" (or "for the current user"). This lets you start SWI-Prolog from the Command Prompt.
5. Finish the installation. On macOS you can alternatively install with Homebrew (`brew install swi-prolog`); on Ubuntu Linux use `sudo apt install swi-prolog`.

D.4.3 Download the project as a ZIP file

1. Open <https://github.com/Suweka/Gemstone-Identification-Assistant> in your browser.
2. Click the green **Code** button, then **Download ZIP**. The file `Gemstone-Identification-Assistant-main.zip` is saved to your Downloads folder.
3. Open the Downloads folder, right-click the ZIP file and choose **Extract All...**, then **Extract**. (On macOS, double-click the ZIP file.)
4. Open the extracted folder `Gemstone-Identification-Assistant-main`, then the `gem_expert` folder inside it. It contains `main.pl`, `knowledge_base.pl`, `engine.pl`, `ui.pl`, `console.pl`, `diagrams.pl`, `examples.pl`, `query_parser.pl`, `tests.pl` and an `images` folder.

If you received the project as a ZIP file in another way (for example by e-mail or on a USB drive), extract it in the same way and open the `gem_expert` folder.

D.4.4 Open the Command Prompt in the project folder

The system is started from a command window opened **inside the `gem_expert` folder**. On Windows, use either method:

- **From File Explorer (easiest):** with the `gem_expert` folder open, click the address bar at the top, type `cmd` and press Enter. A Command Prompt window opens already in the right folder.
- **From the Start menu:** press the Windows key, type `cmd` and press Enter. Then type `cd`, a space, and the folder's location, and press Enter, for example:

```
cd %USERPROFILE%\Downloads\Gemstone-Identification-Assistant-main\gem_expert
```

On macOS or Linux, open Terminal and use `cd` to go to the `gem_expert` folder.

To check that SWI-Prolog is installed, type `swipl --version` and press Enter. You should see a line such as `SWI-Prolog version 10.1.16 for x64-win64`.

If you see "'swipl' is not recognized", SWI-Prolog is not on the PATH. See Troubleshooting (Section D.15).

D.4.5 Start the web interface

1. In the Command Prompt, inside the `gem_expert` folder, type:

```
swipl main.pl
```

A welcome screen explains what the system does and how to use it:

```
=====
GEMSTONE IDENTIFICATION ASSISTANT
=====

An expert system that identifies common Sri Lankan gemstones
from their readings (RI, SG, optic character) and what you can
see (colour, optical effect, inclusions). It tells you whether
a stone is natural, synthetic or an imitation - and explains why.

HOW TO START

1. At the ?- prompt below, type    server(8080).
   (include the full stop) and press Enter.

2. Open your web browser and go to   http://localhost:8080

3. Keep this window open while you use the system.

ON THE HOME PAGE

- Fill in what you know under "Describe your stone".
- IDENTIFY A GEM:  press "Identify stone"
   (forward chaining - works from your inputs to an answer).
- VERIFY A CLAIM:  choose a gem under "Check a claim" and
   press "Verify"  (backward chaining - e.g. "is it a ruby?").
- STEP BY STEP:    press "Start" under "Ask me step by step"
   (the system asks one question at a time).
- New to gems? Try an "Example stone" further down the page.

OTHER COMMANDS (type at the ?- prompt)

start.                use the system here, without a browser
demo.                 show a worked example (blue sapphire)
run_tests.            run the automated tests
stop_server(8080).    stop the web server
halt.                 exit

?-
```

2. At the ?- prompt type the following, **including the full stop**, and press Enter:

```
server(8080).
```

The system replies:

```
% Started server at http://localhost:8080/

The web server is running.

-> Open your web browser and go to   http://localhost:8080
-> Keep this window open while you use the system.
-> To stop: type  stop_server(8080).  then  halt.
```

3. Open your web browser and go to **http://localhost:8080**. The home page appears (Figure D1).
4. On the home page:
 - To **identify** a stone, fill in what you know under **Describe your stone** and press **Identify stone** (Section D.7).
 - To **check a claim** such as "is it a ruby?", choose the gem in **Check a claim** and press **Verify** (Section D.8).
 - To be **guided one question at a time**, press **Start** under **Ask me step by step** (Section D.9).

On Windows the first time you start the server, Windows Defender Firewall may ask whether to allow SWI-Prolog to communicate. Choose **Allow** for private networks.

Every Prolog command ends with a full stop. If nothing happens after pressing Enter, type . and press Enter again.

D.4.6 Stop the system

1. In the Command Prompt type `stop_server(8080).` to stop the web server.
2. Type `halt.` to leave SWI-Prolog. (Closing the Command Prompt window also stops everything.)

D.5 A tour of the home page

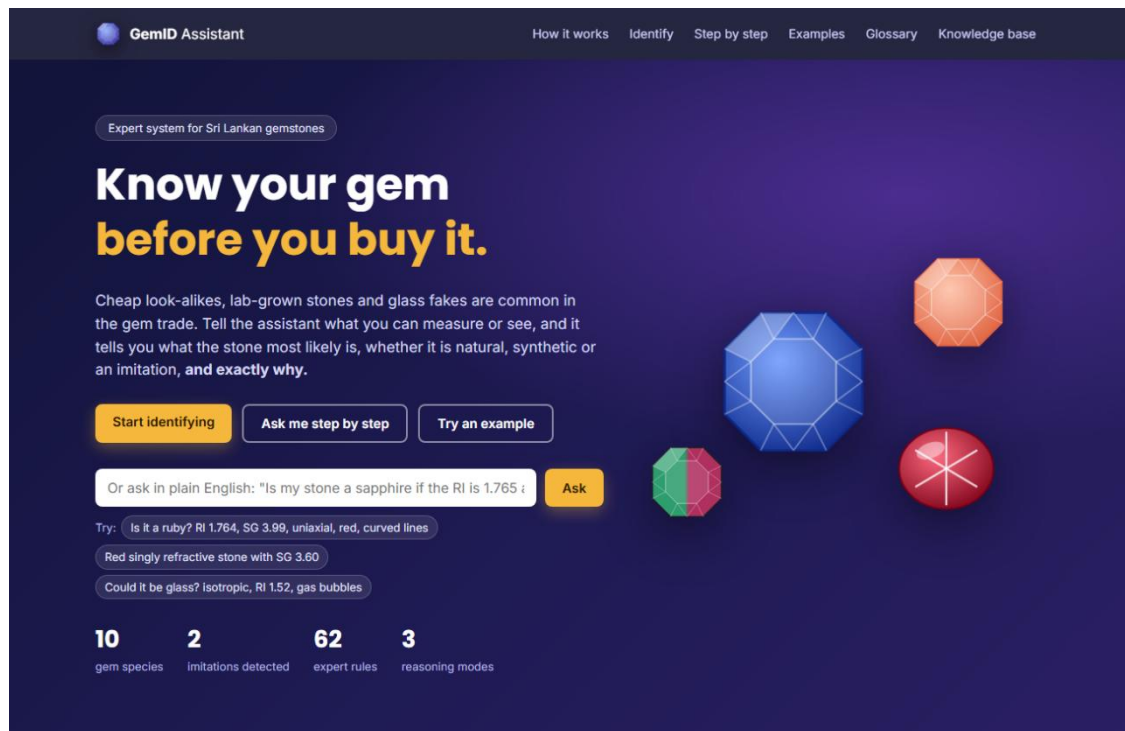


Figure D1. Home page

The top bar links to every part of the system: **How it works**, **Identify** (the input form), **Step by step**, **Examples**, **Glossary** and **Knowledge base**. The introduction offers three buttons: **Start identifying**, **Ask me step by step** and **Try an example**, plus a box for plain-English questions. Scrolling down you will find a summary of what the system tells you, an explanation of how it works and of the two reasoning methods (Figure D2), the input form, example stones and a glossary of gem terms.

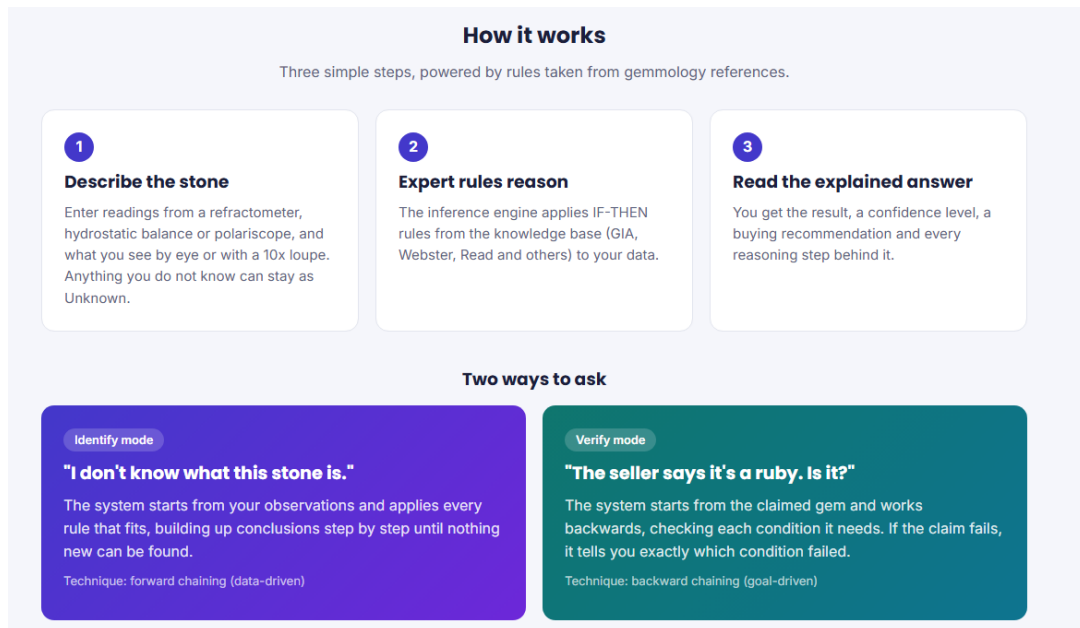


Figure D2. "How it works" section

D.6 Describing your stone

The form (Figure D3) has two groups of fields. **Every field is optional.** Fill in what you know and leave the rest as *Unknown* / *not checked*. The more you fill in, the more certain the answer. Each field shows what the property means (grey text) and how to measure it (**HOW**).

Describe your stone

Every field is optional. The more you fill in, the more confident the answer.

1 Instrument readings

Refractive index (RI)

How strongly the stone bends light. Each gem has its own range.

☐ Over the limit (no reading)

Normal range: gems 1.515 - 1.990 (with imitations 1.440 - 2.180). Most refractometers read up to about 1.81. [See the range for each gem](#)

HOW Read it from a gem refractometer (use the highest reading).

Specific gravity (SG)

How heavy the stone is compared with the same volume of water.

Normal range: gems 2.55 - 4.80 (with imitations 2.20 - 6.00). [See the range for each gem](#)

HOW Weigh the stone in air and in water (hydrostatic balance).

2 What you can see

Colour

The main body colour of the stone. Pick the closest swatch.

☐ Not checked
 ☐ Red
 ☐ Pink
 ☐ Orange
 ☐ Yellow
 ☐ Green
 ☐ Blue
 ☐ Purple
 ☐ Brown
 ☐ Black
 ☐ White
 ☐ Colourless

HOW Look at it in daylight against a white background.

Optical effect

Special light effects. Pick the picture that looks like your stone.

☐ Not checked
 ☐ None
 ☐ Star
 ☐ Cat's-eye
 ☐ Colour change
 ☐ Moonstone glow

HOW Look under a single light source and rotate the stone.

Inclusions

Tiny features inside the stone that reveal how it formed. Pick the sketch closest to what you see.

☐ Not checked
 ☐ Clean
 ☐ Silk
 ☐ Crystals
 ☐ Fingerprints
 ☐ Gas bubbles
 ☐ Curved lines
 ☐ Swirl marks
 ☐ Discoid fractures

HOW Examine it with a 10x jeweller's loupe.

Identify my stone

Find the best match using every rule in the knowledge base.

Check a claim

Test whether the stone could be one particular gem.

Ask me step by step

The system asks only the questions it needs, and you can ask it why.

[Clear the form](#)

Figure D3. The input form: instrument readings with normal ranges, and picture choices for the optic character and for what you can see

Table D3. Input fields

Field	What it is	How to obtain it	Tips
Refractive index (RI)	How strongly the stone bends light. Each gem has its own range.	Gem refractometer with contact liquid; use the highest reading.	Enter three decimals, e.g. 1 . 765. Allowed range 1.3-3.0. If the scale shows no shadow edge at all, tick Over the limit (no reading) instead. Leave it blank if you have no refractometer.
Specific gravity (SG)	How heavy the stone is compared with the same volume of water.	Hydrostatic balance: $SG = \text{weight in air} / (\text{weight in air} - \text{weight in water})$.	Enter two decimals, e.g. 4 . 00. Allowed range 1.0-8.0.
Optic character	Whether light passes as one ray (singly refractive) or is split in two (doubly refractive).	Polariscope: rotate the stone between crossed filters. A conoscope (or a glass sphere held over the stone) shows the interference figure.	Click the picture that matches what you see: <i>Stays dark</i> (singly refractive), <i>Blinks light/dark</i> (doubly refractive, figure not checked), <i>Dark cross</i> (uniaxial) or <i>Curved bars</i> (biaxial).
Colour	The main body colour.	Look in daylight against a white background.	Click the swatch closest to your stone.
Optical effect	Star, cat's-eye, colour change or moonstone glow.	Look under a single light source and rotate the stone; compare daylight and lamp light for colour change.	Click the picture that looks like your stone. Choose <i>None</i> if you looked and saw no effect.
Inclusions	Tiny features inside the stone that reveal how it formed.	Examine with a 10x jeweller's loupe.	Click the sketch closest to what you see. <i>Clean</i> means you looked and saw nothing inside.

Optic character, colour, optical effect and inclusions are chosen by clicking a **picture tile**; the chosen tile is outlined in blue. Leave the **Not checked** tile selected if you did not look. Hover over a tile to see its full description.

D.6.1 Instant checks on your readings

As soon as you type an RI or SG value, a short note appears under the box (Figure D4):

- **Green "Fits: ..."** lists the gems whose range includes your reading.
- **Orange warning:** the value is unusual (no gem in the system has it, or it is above what most refractometers can read). Check your measurement.
- **Red error:** the value is impossible (outside the allowed range). It appears when you leave the box, and the form will not be sent until you correct it.

Under each box a **Normal range** line shows the range of readings covered by the system (for example RI 1.515 - 1.990 for gems). The link **See the range for each gem** opens the ranges of every gem in a new tab, so your form is kept.

Ticking **Over the limit (no reading)** greys out the RI box and shows which gems read above the refractometer's limit (zircon, cubic zirconia and some garnets).

1 Instrument readings

Refractive index (RI)
How strongly the stone bends light. Each gem has its own range.

1.72

✓ Fits: Spinel, Glass.

☐ Over the limit (no reading)

Normal range: gems 1.515 - 1.990 (with imitations 1.440 - 2.180). Most refractometers read up to about 1.81. [See the range for each gem](#)

HOW Read it from a gem refractometer (use the highest reading).

Specific gravity (SG)
How heavy the stone is compared with the same volume of water.

9

⚠ SG must be between 1.0 and 8.0.

Normal range: gems 2.55 - 4.80 (with imitations 2.20 - 6.00).
[See the range for each gem](#)

HOW Weigh the stone in air and in water (hydrostatic balance).

1 Instrument readings

Refractive index (RI)
How strongly the stone bends light. Each gem has its own range.

e.g. 1.765

✓ Over the limit: fits Garnet, Zircon, Cubic zirconia.

☒ Over the limit (no reading)

Normal range: gems 1.515 - 1.990 (with imitations 1.440 - 2.180). Most refractometers read up to about 1.81. [See the range for each gem](#)

HOW Read it from a gem refractometer (use the highest reading).

Specific gravity (SG)
How heavy the stone is compared with the same volume of water.

1.9

⚠ Unusual reading: no gem in the knowledge base has an SG of 1.9. You can still continue, but please check the reading.

Normal range: gems 2.55 - 4.80 (with imitations 2.20 - 6.00).
[See the range for each gem](#)

HOW Weigh the stone in air and in water (hydrostatic balance).

Figure D4. Readings checked as you type (top) and the "Over the limit" option (bottom)

Below the fields are three action panels: **Identify my stone**, **Check a claim** (choose a gem in the list, then **Verify**) and **Ask me step by step**. **Clear the form** empties all fields.

D.7 Identifying a stone

D.7.1 Steps

1. Fill in the fields you know (Section D.6).
2. Press **Identify stone**.
3. Read the result page (Figure D5).

D.7.2 Reading the result

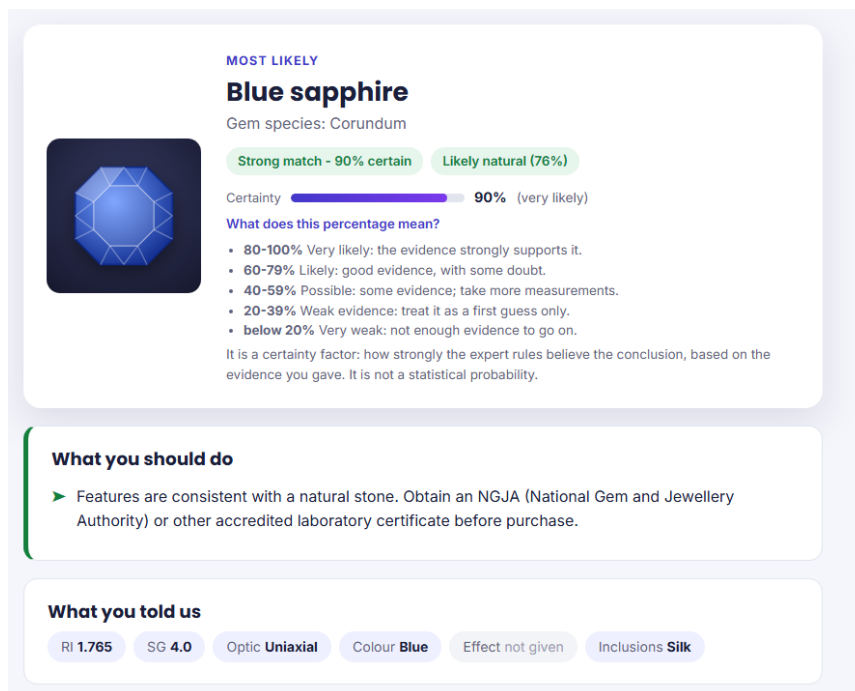


Figure D5. Result for a natural blue sapphire, with the certainty scale opened

Table D4. Parts of the result page

Part	Meaning
Title (e.g. <i>Blue sapphire</i>)	The most likely identification, with the species underneath.
Confidence badge	Strong match (green): optic character, RI and SG all agree. Possible match (amber): data is incomplete or fits more than one gem. No match (grey): nothing in the knowledge base fits.
Certainty bar	How strongly the expert rules believe the conclusion, with a word that explains it: very likely (80-100%), likely (60-79%), possible (40-59%), weak evidence (20-39%), very weak (below 20%). Click What does this percentage mean? to see the scale. It is a certainty factor, not a statistical probability.
Origin badge	Likely natural (green), Synthetic or laboratory test recommended (amber), Imitation or Glass or synthetic (red), Not determined (grey, usually because no inclusions were reported). A Probably heat-treated badge appears when heat indicators were seen.
What you should do	Practical recommendations.
What you told us	Your inputs; fields you left empty show "not given".
Reasoning diagram	A picture of how the rules fired (Section D.7.3).
How the system reached this	Each reasoning step in plain English, with the rule number, its certainty, the expert knowledge behind it and its source.

Part	Meaning
About (gem)	Reference facts about the identified gem.
Technical trace	Click to open the full list of facts and fired rules (for gemmologists and examiners).
Buttons	Edit my inputs returns to the form with your values filled in; Print / Save as PDF prints the result (Section D.12); Start a new stone clears the form.

D.7.3 Using the reasoning diagram



Figure D6. Forward-chaining diagram

Your inputs are the blue boxes at the top. Each dark oval is a rule; its gold number shows the order in which it fired. Arrows show which facts made the rule fire and which new fact it added.

Use the buttons:

- **Play step by step:** replays the reasoning automatically, one rule at a time, with a caption explaining each step.
- **Back / Next:** move one step at a time yourself.
- **Show all:** shows the complete diagram again.

Hover the mouse over any box to see its full details.

D.7.4 When some data is missing

If you leave out the RI, SG or optic character, the system cannot confirm a species. Instead it lists **Ranked possibilities** (Figure D7): every gem consistent with what you entered, ordered by certainty, with the evidence for each.

Above the list, the yellow **Most useful next measurement** box tells you which missing measurement would narrow the list down best, and what each candidate would read. For example, for a red, singly refractive stone it advises measuring the refractive index, because spinel reads 1.712-1.736 and garnet 1.730-1.890.

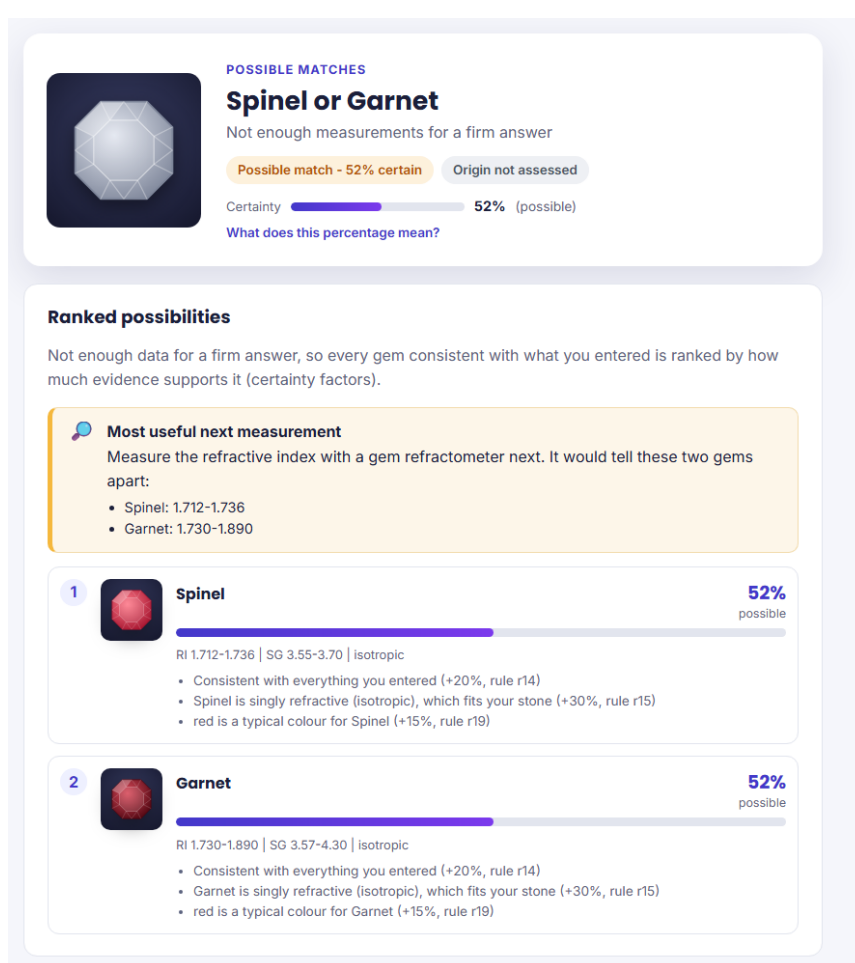


Figure D7. Ranked possibilities and the most useful next measurement for a red, singly refractive stone

D.8 Checking a claim (Verify)

1. Fill in the fields you know.
2. In the **Check a claim** panel choose the gem the seller claims, e.g. *Ruby*.

3. Press **Verify**.

The answer is a clear **Yes** (green tick) or **No** (red cross). For a **No**, the page shows the **main reason**, what your data points to instead, and a **Why not?** list tracing the failure back through the rules (Figure D8). The backward-chaining diagram shows the question at the top, the rules tried below it, and each condition marked proved (green), failed (red) or never checked (dashed grey).

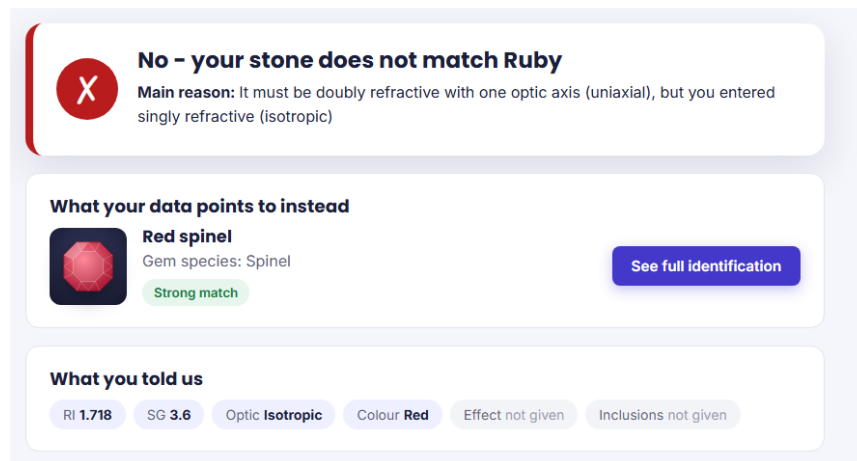


Figure D8. A failed verification: a red spinel offered as a ruby

D.9 Step-by-step consultation

1. Click **Step by step** in the top bar (or **Ask me step by step** on the home page).
2. Choose what you want to find out. *Find the species* lets the system test one gem hypothesis after another. You can also choose a particular gem.
3. Answer each question and press **Answer**, or press **I don't know**. The bar at the top shows how far you are ("Question 2 of at most 6").
4. Made a mistake? Press **Back** to return to the previous question and change your answer.
5. Open **Why are you asking this?** at any time to see which rule the system is testing and why it needs the answer (Figure D9).
6. When the system has enough information it shows the answer, how many questions it needed, and a diagram of its reasoning. **Change my last answer** takes you back one step.

Question 2 of at most 6

QUESTION 2 testing rule r1

What is the refractive index (RI) of your stone?

Refractive index (RI)
How strongly the stone bends light. Each gem has its own range.

☐ Over the limit (no reading)

Normal range: gems 1.515 - 1.990 (with imitations 1.440 - 2.180). Most refractometers read up to about 1.81.
[See the range for each gem](#)

HOW Read it from a gem refractometer (use the highest reading).

Answer **I don't know** [← Back](#)

▼ **Why are you asking this?**

1. I am trying to find out: is it Ruby? Rule r21 says: IF Species: Corundum AND Colour: Red AND NOT Effect: Star THEN Variety: Ruby.
2. To use that rule I first need to establish: is it Corundum? Rule r1 says: IF Optic: Uniaxial AND RI in Corundum range AND SG in Corundum range THEN Species: Corundum.
3. To test rule r1 I need to know: the refractive index.

Figure D9. A consultation question with the progress bar, the Back button and the explanation

The system asks only what it needs. For example, when checking for ruby, answering *Isotropic* to the first question ends the consultation immediately, because a ruby must be doubly refractive.

D.10 Asking in plain English

Type a question into the box on the home page and press **Ask**. The system shows what it understood (Figure D10) and then answers. Examples:

- *Is my stone a sapphire? RI 1.765, SG 4.00, uniaxial and blue*
- *What is a singly refractive red stone with SG 3.60?*
- *Could it be glass? isotropic, RI 1.52, gas bubbles*
- *Blue stone, RI over the limit, SG 4.70, uniaxial*

Understood words include RI / refractive index, "over the limit", SG / specific gravity, singly or doubly refractive, uniaxial, biaxial, colour names, star, cat's eye, colour change, adularescence, silk, crystals, fingerprints, gas bubbles, curved lines, swirl marks, discoid fractures, clean, and gem names. Words the system did not use are listed so you can rephrase.

You asked

"Is my stone a sapphire? RI 1.765, SG 4.00, uniaxial and blue"

The grammar (a Prolog DCG) understood your question as:

Question **Is it Blue sapphire?** RI **1.765** SG **4.0** Optic **Uniaxial** Colour **Blue**

Yes – your stone is consistent with Blue sapphire

Every condition the expert rules require for Blue sapphire is met by what you entered.

Certainty **90%** (very likely)

[What does this percentage mean?](#)

A lab certificate is still recommended before buying.

Figure D10. A plain-English question and how it was understood

D.11 Worked examples

The **Example stones to try** section of the home page has 15 ready-made stones (Figure D11). Press **Try it** on any card to copy its readings into the form, then press **Identify stone** or **Verify**.

Example stones to try

Press "Try it" to load a stone's readings into the form, then choose Identify or Verify.

 Blue sapphire RI 1.765 SG 4.00 Optic Uniaxial Colour Blue Inclusion Silk EXPECTED Blue sapphire, likely natural Try it	 Synthetic ruby RI 1.764 SG 3.99 Optic Uniaxial Colour Red Inclusion Curved growth lines EXPECTED Ruby, synthetic (flame fusion) Try it	 Star sapphire RI 1.765 SG 4.00 Optic Uniaxial Colour Blue Effect Star (asterism) Inclusion Silk EXPECTED Star sapphire, likely natural Try it	 Padparadscha RI 1.766 SG 4.00 Optic Uniaxial Colour Orange Inclusion Fingerprints / healed fractures EXPECTED Padparadscha, likely natural Try it
			

Figure D11. Example stones

To test every gem in the system with your own inputs, use the readings in **Section D.19**: each gem's facts, a sample reading to type, and the result the system gives.

D.11.1 Example 1: a natural blue sapphire

Input	RI 1.765, SG 4.00, Uniaxial, Blue, Silk
Result	Blue sapphire (species corundum) - Strong match, 90% certain (very likely); origin Likely natural (76%) .
Recommendation	Obtain an NGJA or other accredited laboratory certificate before purchase.
Why	Rule r1 identified corundum from the optic character, RI and SG; rule r24 named the blue variety; rule r46 used the rutile silk as evidence of natural origin.

D.11.2 Example 2: a glass imitation

Input	RI 1.60, Isotropic, Gas bubbles (SG not measured)
Result	Imitation: Glass - 85% certain; origin Imitation (not a natural gemstone) (Figure D12).
Recommendation	This is an imitation with little gem value; do not pay gemstone prices.

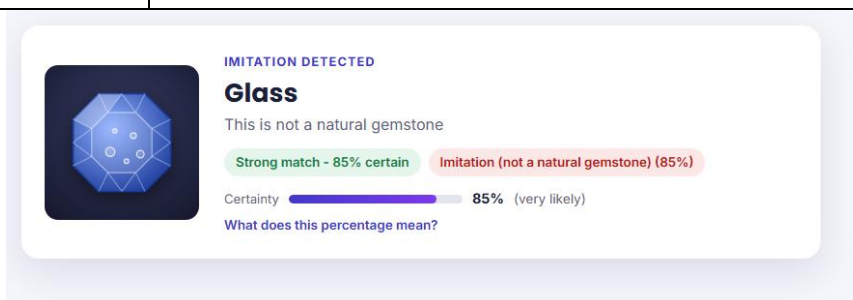


Figure D12. A glass imitation is detected

D.11.3 Example 3: checking a claim that fails

Input	RI 1.718, SG 3.60, Isotropic, Red; claim: <i>Ruby</i>
Result	No - the stone does not match ruby. Main reason: a ruby must be doubly refractive (uniaxial) but the stone is singly refractive. The data points to Red spinel instead (Figure D8).

D.11.4 Example 4: a refractometer reading over the limit

Input	RI Over the limit (ticked), SG 4.70, Uniaxial, Blue, Crystals
Result	Blue zircon (species zircon) - Strong match, 65% certain (likely); origin Likely natural (Figure D13).
Why	Only zircon, cubic zirconia and some garnets read above the refractometer's limit; of those, only zircon is uniaxial with an SG of about 4.7. The certainty is lower than for a measured RI because the exact value is unknown.

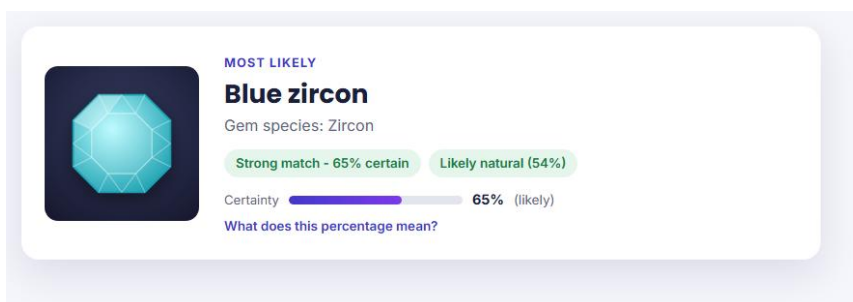


Figure D13. A blue zircon identified with an RI over the refractometer limit

D.12 Printing or saving a result

Every result page has a **Print / Save as PDF** button at the bottom. It opens your browser's print window: choose a printer, or choose **Save as PDF** as the printer to keep a file. The printout (Figure D14) has a header with the date and time and a reminder that it is a first opinion, leaves out the menu and buttons, and includes the full diagram and technical trace. You can take it to a seller or a gem laboratory.

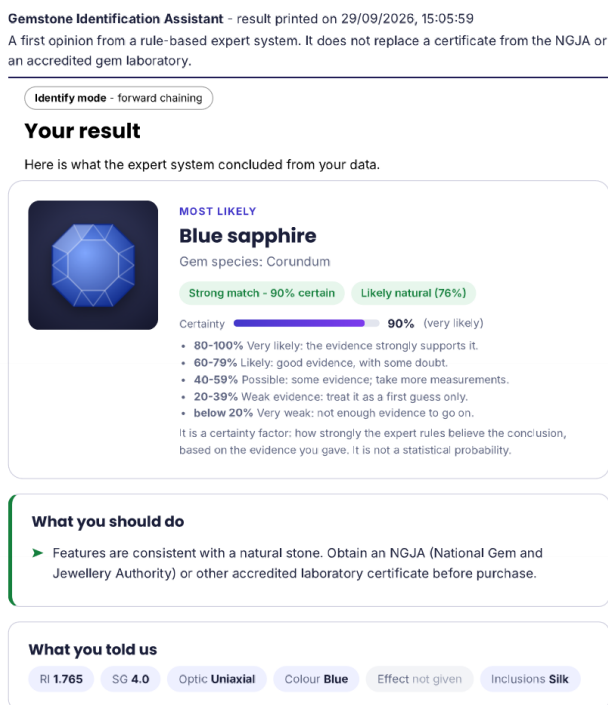


Figure D14. First page of a printed result

D.13 Your recent stones

The system remembers the last eight results you looked at. They appear at the top of the form on the home page under **Your recent stones** (Figure D15). Click one to open its result again, or press **Clear list** to remove them.

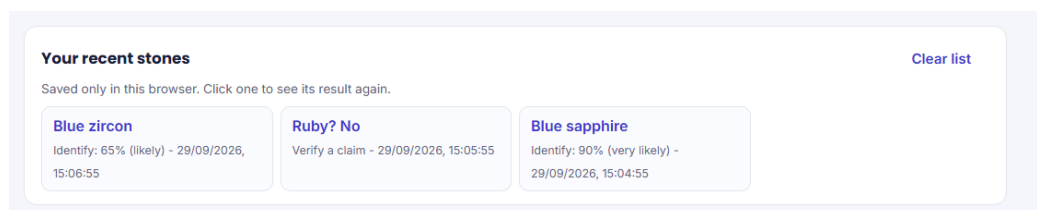


Figure D15. Your recent stones on the home page

The list is stored only in your own browser on your computer; nothing is sent anywhere. It will be empty in a private window, in a different browser, or after clearing your browsing data.

D.14 Using the system on a phone or tablet

The pages adapt to small screens (Figure D16). On a narrow screen the links in the top bar are replaced by a **menu button** (three lines) in the top right corner: tap it to open the menu, and tap a link or press it again to close. The easiest way to use a phone or tablet is the online version: open **<https://gemstone-identification-assistant.onrender.com>** in its browser. Alternatively, to use a phone on the same network as a computer running the system, open `http://<computer's IP address>:8080` on the phone; Windows Firewall must allow SWI-Prolog on private networks.

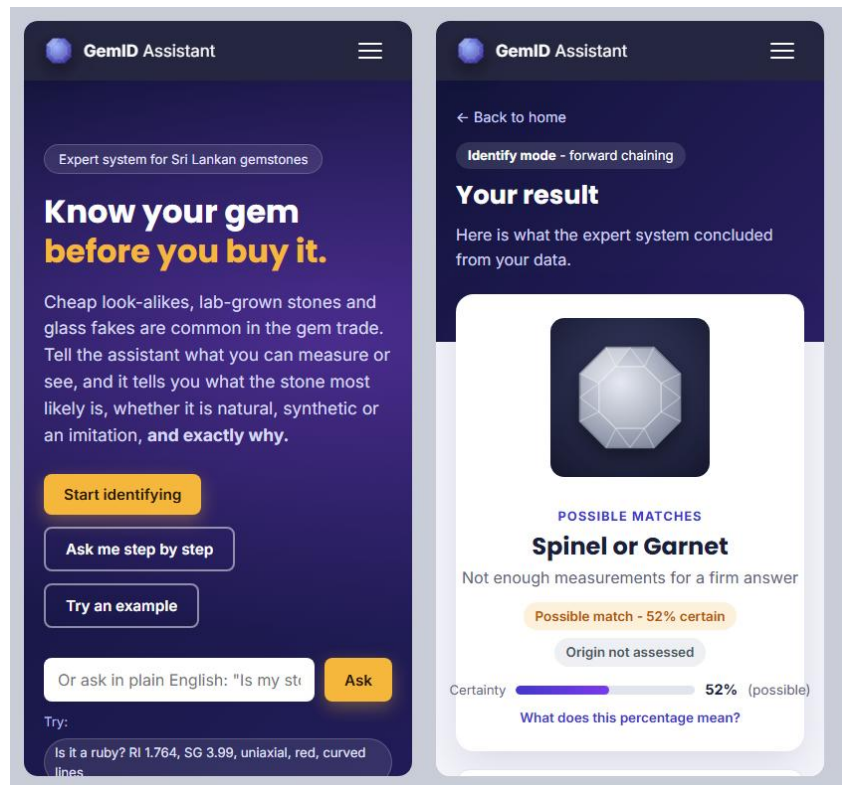


Figure D16. Home page and a result on a phone-sized screen

D.15 Troubleshooting

Table D5. Common problems and solutions

Problem	Solution
"swipl" is not recognized as an internal or external command"	SWI-Prolog is not on the PATH. Re-run the installer and select "Add swipl to the system PATH", then open a new terminal. Alternatively run it with its full path: "C:\Program Files\swipl\bin\swipl.exe" main.pl.
Error "port 8080 is already in use" (or similar) after <code>server(8080) .</code>	Another program uses that port. Start on a different port, e.g. <code>server(8081) .</code> , and open <code>http://localhost:8081</code> .
Browser shows "This site can't be reached"	The server is not running. Check that you typed <code>server(8080) .</code> with the full stop, and that the terminal window is still open. Check the port number in the address.
Nothing happens after typing a command	Prolog is waiting for the full stop. Type <code>.</code> and press Enter.
Error messages about files not found when running <code>swipl main.pl</code>	The terminal is not in the <code>gem_expert</code> folder. Use <code>cd</code> to move into it, or open the terminal from that folder.

Problem	Solution
"Please check your input" page, or a red message under a box	A value is outside the allowed range (RI 1.3-3.0, SG 1.0-8.0) or is not a number. Correct it (or press Go back and fix it) and try again.
The RI box is greyed out	Over the limit (no reading) is ticked. Untick it to type an RI value.
"Your recent stones" is missing or empty	It only appears once you have looked at a result in this browser. It is empty in private windows, in another browser, or after clearing browsing data.
The print button prints extra pages or no colours	In the print window, turn on Background graphics and choose Save as PDF or your printer; the result normally fits on a few pages.
Result says "We could not identify this stone"	Re-check the readings (RI usually has three decimals, SG two) and the optic character. The stone may also be outside the system's scope; seek a laboratory test.
Changes to the program files do not appear	In the Prolog terminal type <code>make .</code> to reload changed files, then refresh the browser (Ctrl+F5). Or stop and restart the system.
Windows Firewall message on first start	Choose Allow for private networks.
The online version is slow to open, or shows a loading or error page	The free hosting service sleeps after about 15 minutes without visitors. Wait up to a minute and refresh the page. If it still does not load, use the system on your own computer (Section D.4).
No Download ZIP option on GitHub, or the page is not found	The repository may be private. Ask the author for access or for the ZIP file, then continue from Section D.4.3 step 3.
The page fonts look plain	You are offline; the page falls back to standard fonts. Everything still works.

D.16 Console mode (without a browser)

When the system runs on your own computer (Option B, Section D.4), the same functions are also available in the Command Prompt, without a browser. This mode is not available in the online version.

Step 1. Start SWI-Prolog with `swipl main.pl`.

Step 2. Type `start`. A menu appears:

```
What would you like to do?
```

1. Identify a stone - "I don't know what it is" (forward chaining)
2. Check a claim - "The seller says it's a ruby" (backward chaining)
3. Step by step - asks only what it needs; type WHY at any question
4. Ask in English - e.g. is it a ruby if the RI is 1.765 and it is red
5. Run the demo - natural blue sapphire example
6. Quit

```
Choose 1-6:
```

Step 3. Type a number and press Enter (no full stop is needed inside the menu). Answer each question by typing a number or a value; press Enter on its own if you do not know. For the RI, type OTL if the refractometer shows no reading (over the limit).

Step 4. In step-by-step mode, type `why` at any question to see why it is being asked:

```
What is the optic character?
```

0. Don't know
1. Isotropic
2. Uniaxial
- ...

```
> why
```

```
WHY I AM ASKING:
```

- I am trying to find out: is it Corundum? Rule r1 says: IF Optic: Uniaxial AND RI in Corundum range AND SG in Corundum range THEN Species: Corundum.
- To test rule r1 I need to know: the optic character.

Step 5. After each result you can type `y` to see the technical trace.

Step 6. Choose 6 to leave the menu, then type `halt`.

The command `demo.` runs the worked blue sapphire example directly.

D.17 Running the automated tests

Step 1. Start SWI-Prolog with `swipl main.pl`.

Step 2. Type `run_tests.` The last line should read:

```
% All 47 tests passed
```

The rule base can also be exported to a spreadsheet file with `export_rules_csv('rules_table.csv').`

D.18 Quick reference

Table D6. Commands

Command (type at the ?- prompt)	Effect
swipl main.pl	(in the terminal) Load the system.
server(8080).	Start the web interface at http://localhost:8080 .
stop_server(8080).	Stop the web interface.
start.	Start the console interface.
demo.	Run the worked blue sapphire example.
run_tests.	Run the 47 automated tests.
make.	Reload program files after editing them.
halt.	Leave SWI-Prolog.

Table D7. Web pages

Address	Page
https://gemstone-identification-assistant.onrender.com	Online version (no installation)
http://localhost:8080/	Home page, form, examples, glossary (on your own computer)
http://localhost:8080/consult	Step-by-step consultation
http://localhost:8080/rules	Knowledge base: gem facts and all rules

D.19 Gem property facts for testing

This table lists the facts the system holds for every material in its knowledge base, so the system can be tested with any gem. For each one:

- **RI** and **SG** are the reading ranges the rules use. The system also accepts readings slightly outside them ($RI \pm 0.005$, $SG \pm 0.03$) to allow for measurement error, but with lower certainty.
- **Try this reading** is a sample input from the middle of the ranges. Enter it in the form (Section D.6) and press **Identify stone**. For "RI over the limit", tick **Over the limit (no reading)** instead of typing an RI.
- **System result** is what the system concludes for that reading, with its certainty. These results were produced by running the system on each sample.

- **Source** is the reference the values were taken from (full references are in the References section of this report).

Adding a **colour** from the list names the variety (for example corundum + red gives *Ruby*, chrysoberyl + colour change gives *Alexandrite*), and adding an **inclusion** lets the system judge whether the stone is natural or synthetic. Trying values *outside* the ranges, or a different optic character, shows how the system rejects a gem and explains why.

Table D8. Gem property facts, sample readings and results

	Material and colours	RI	SG	Optic	Mohs	Try this reading	System result	Source
	Corundum Red, Pink, Orange, Yellow, Green, Blue, Purple, Colourless	1.757 - 1.780	3.95 - 4.10	Uniaxial	9	RI 1.769, SG 4.03, Uniaxial	Corundum (95%)	GIA Gem Encyclopedia
	Spinel Red, Pink, Orange, Blue, Purple	1.712 - 1.736	3.55 - 3.70	Isotropic	8	RI 1.724, SG 3.63, Isotropic	Spinel (90%)	Webster, Gems
	Garnet Red, Orange, Green, Purple, Brown	1.730 - 1.890	3.57 - 4.30	Isotropic	7	RI 1.770, SG 3.93, Isotropic	Garnet (90%)	Webster, Gems
	Chrysoberyl Yellow, Green, Brown	1.740 - 1.760	3.70 - 3.78	Biaxial	8.5	RI 1.750, SG 3.74, Biaxial	Chrysoberyl (95%)	GIA Gem Encyclopedia
	Zircon Red, Orange, Yellow, Green, Blue, Brown, Colourless	1.810 - 1.990	4.60 - 4.80	Uniaxial	7.5	RI over the limit, SG 4.70, Uniaxial	Zircon (77%)	Gemdat.org
	Topaz Pink, Orange, Yellow, Blue, Brown, Colourless	1.606 - 1.644	3.49 - 3.57	Biaxial	8	RI 1.625, SG 3.53, Biaxial	Topaz (95%)	GIA Gem Encyclopedia
	Tourmaline Red, Pink, Yellow, Green, Blue, Brown, Black	1.614 - 1.666	3.00 - 3.26	Uniaxial	7.5	RI 1.640, SG 3.13, Uniaxial	Tourmaline (90%)	GIA Gem Encyclopedia

	Material and colours	RI	SG	Optic	Mohs	Try this reading	System result	Source
	Quartz Pink, Yellow, Purple, Brown, Colourless	1.540 - 1.558	2.60 - 2.70	Uniaxial	7	RI 1.549, SG 2.65, Uniaxial	Quartz (90%)	Read, Gemmology
	Feldspar White, Colourless	1.515 - 1.530	2.55 - 2.63	Biaxial	6	RI 1.523, SG 2.59, Biaxial	Feldspar (90%)	Read, Gemmology
	Beryl Pink, Yellow, Green, Blue, Colourless	1.564 - 1.602	2.63 - 2.92	Uniaxial	7.5	RI 1.583, SG 2.78, Uniaxial	Beryl (90%)	Gemdat.org
	Glass Made in any colour	1.440 - 1.770	2.20 - 4.50	Isotropic	5.5	RI 1.605, SG 3.35, Isotropic, gas bubbles	Glass (85%)	International Gem Society
	Cubic zirconia Usually colourless; also dyed colours	2.150 - 2.180	5.60 - 6.00	Isotropic	8.5	RI over the limit, SG 5.80, Isotropic	Cubic zirconia (95%)	GIA Gem Encyclopedia

These values are for testing the system. Real stones vary; always rely on a certificate from the NGJA or an accredited laboratory for a purchase.